

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

Practical - VII

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Class: TYIT	Batch: 2
Date of Assignment: 21-8-26	Date/Time of Submission: 21-8-26

Creating Interactive Dashboards and Data Stories : -

A] Design a basic dashboard layout using sheets :

Sheet 1 — Matches Won by Team

This follows the PDF directly.

1. Click **Sheet 1**.
2. Rename it:

Matches Won by Team

3. From matches, drag **winner** → **Rows**.
4. Drag **id** → **Columns**.
5. Click the CNT (id) pill if necessary and select **Count (Distinct)**.
6. Drag **winner** → **Filters**.
7. Remove **Null**.
8. Click **OK**.
9. Click **Sort Descending**.
10. Drag **winner** → **Color**.
11. Click **Label**.
12. Tick **Show mark labels**.



Name deliveries.csv

How do relationships differ from joins? [Learn more](#)

matches.csv

Id

Operator

=

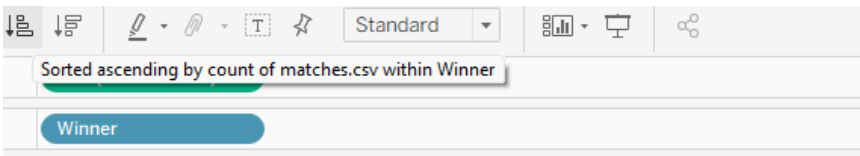
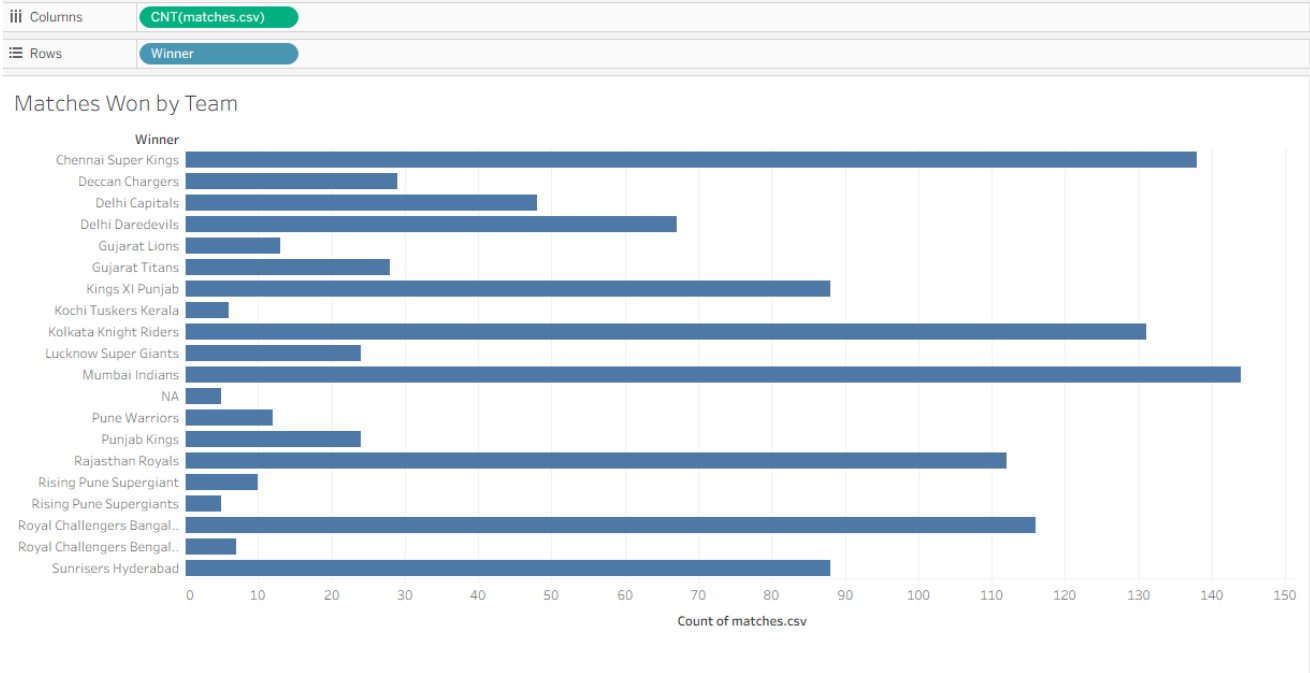
deliveries.csv

Match Id

Add more fields

> Performance Options

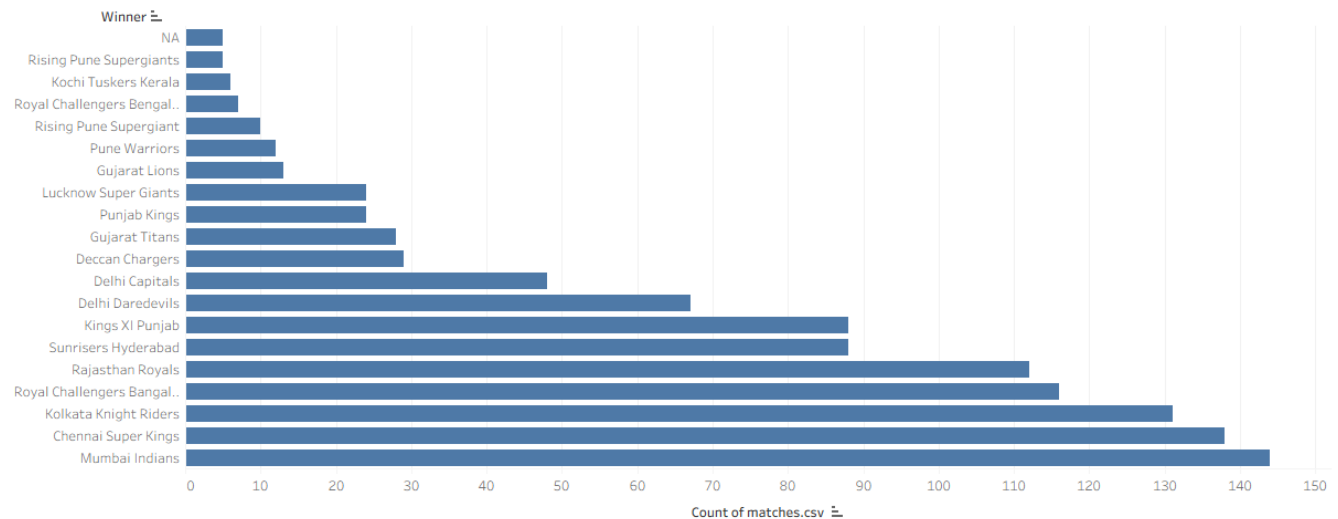
Remove Relationship



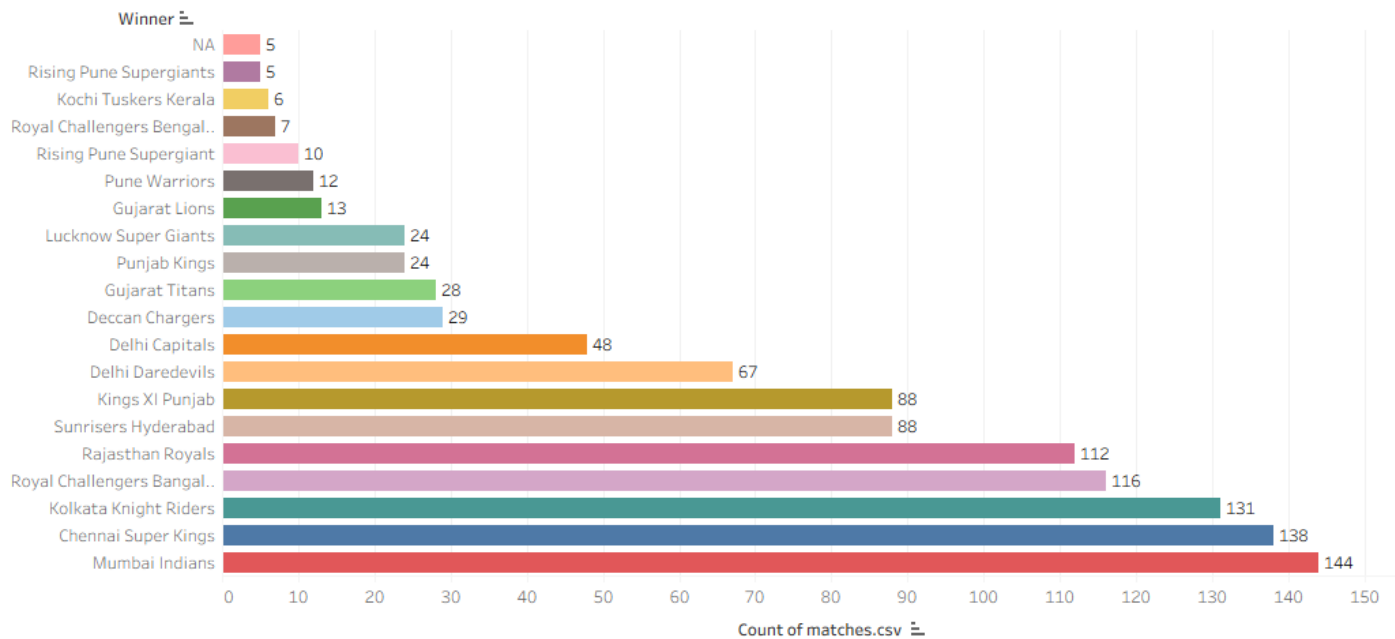
Won by Team



Matches Won by Team



Matches Won by Team T032



Result:

A bar chart showing the number of matches won by each IPL team.

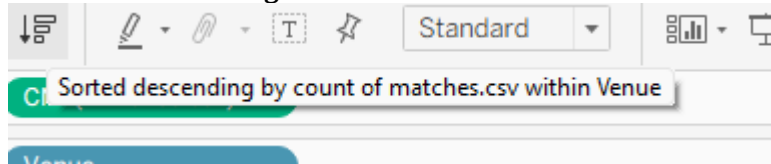
Sheet 2 — Top Venues :

Again, this follows the PDF.

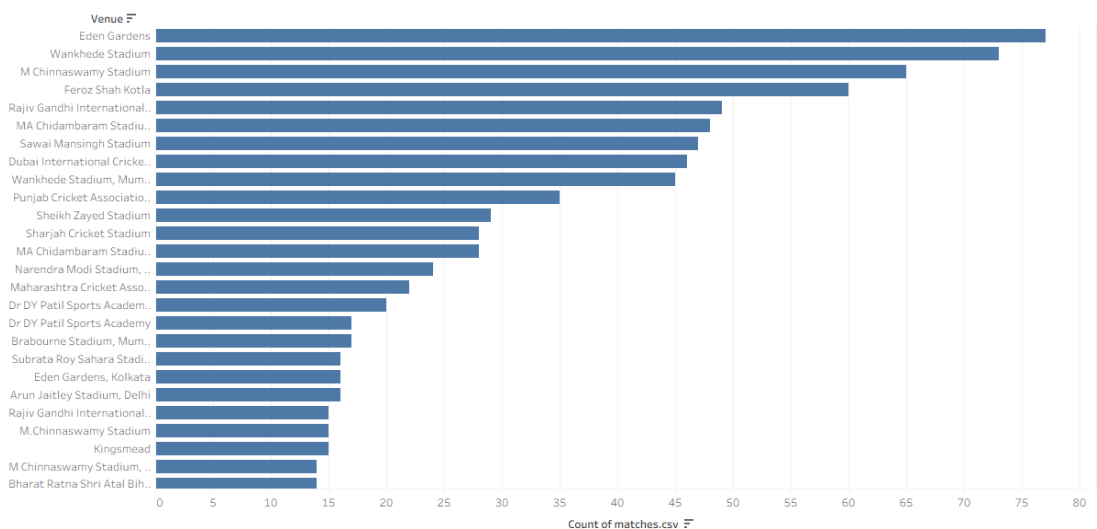
1. Click **New Worksheet**.
2. Rename it:

Top Venues

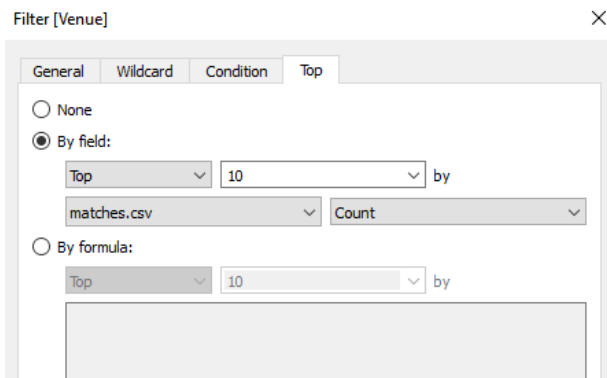
3. Drag **venue** → **Rows**.
4. Drag **id** → **Columns**.
5. Use **COUNTD(id)**.
6. Right-click **venue** → **Sort**.
7. Select **Descending**.



Top Venues T032



8. Drag **venue** → **Filters**.



9. Click the **Top** tab.

10. Select **By field**.

11. Set:

Top 10 by COUNTD(id)

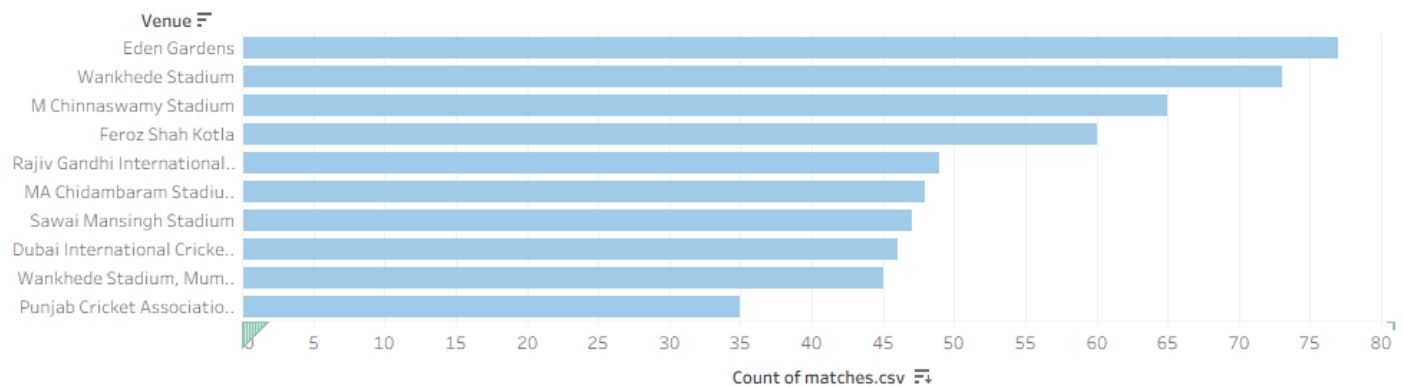
12. Click **OK**.

13. On the Marks card select **Bar**.

Result:

Top Match Venues

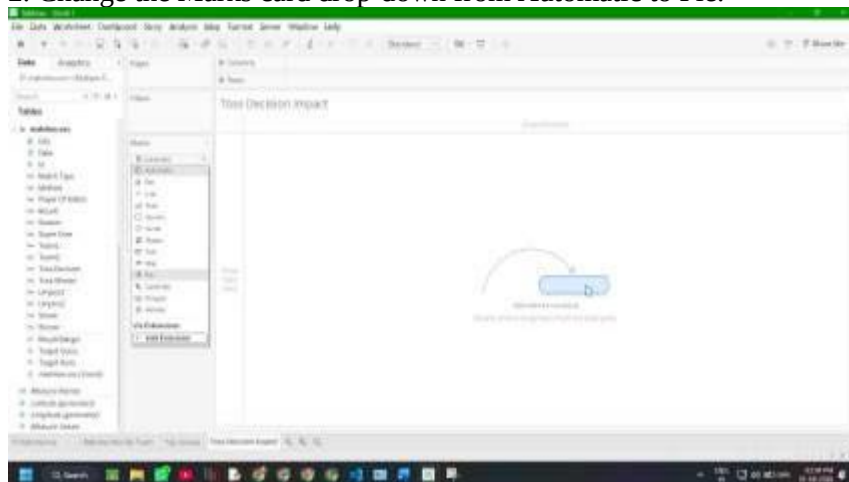
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Step 3: Create Sheet 3 "Toss Decision Impact" (Pie Chart or Donut Chart)

1. Open a new worksheet and rename it: Toss Decision Impact.

2. Change the Marks card drop-down from Automatic to Pie.



3. Drag Toss Decision (bat/field) to Color on the Marks card.

4. Drag Matches (Count) to Angle on the Marks card.

5. Drag Toss Decision and Matches (Count) to Label on the Marks card

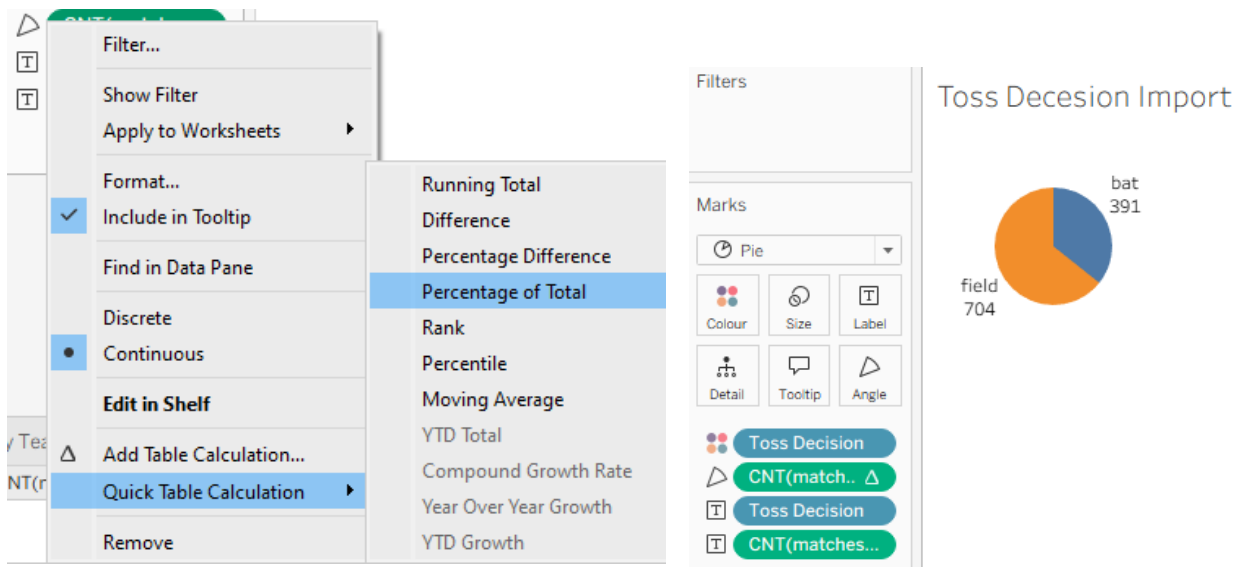
6. Look at the Marks card on the left side of your chart view.

7. Find the green pill named CNT(Matches) (or CNTD(Id)) that has a small T or Label icon next to it.

8. Right-click directly on that green pill (or click its small drop-down arrow).

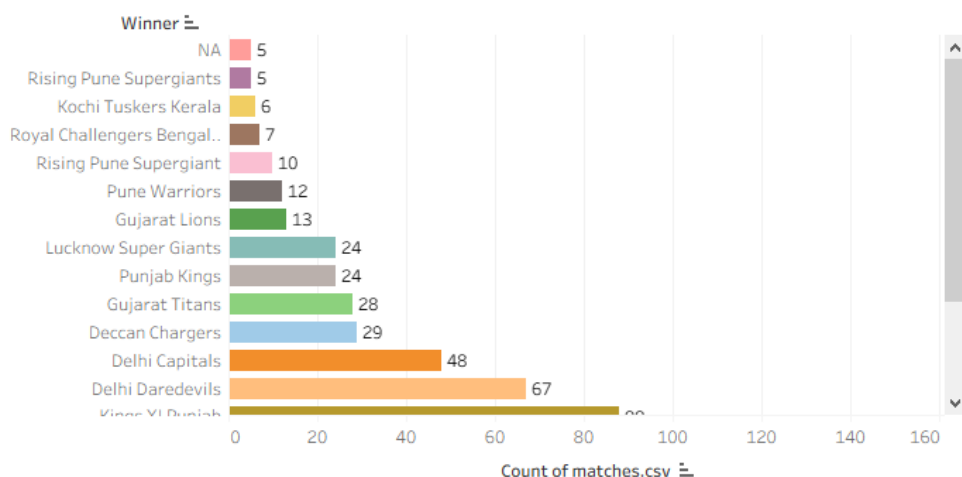
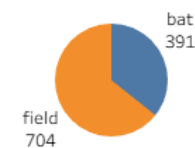
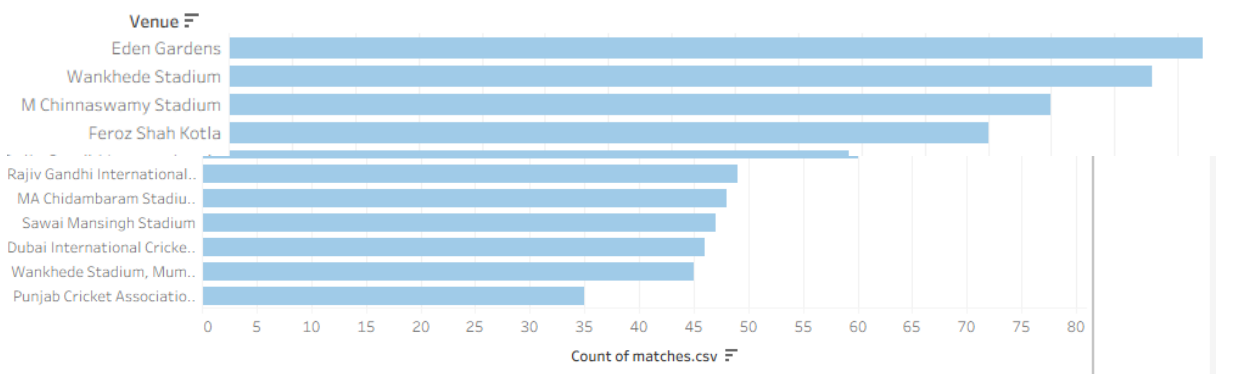
9. In the pop-up menu, hover over Quick Table Calculation.

10. Select Percent of Total.



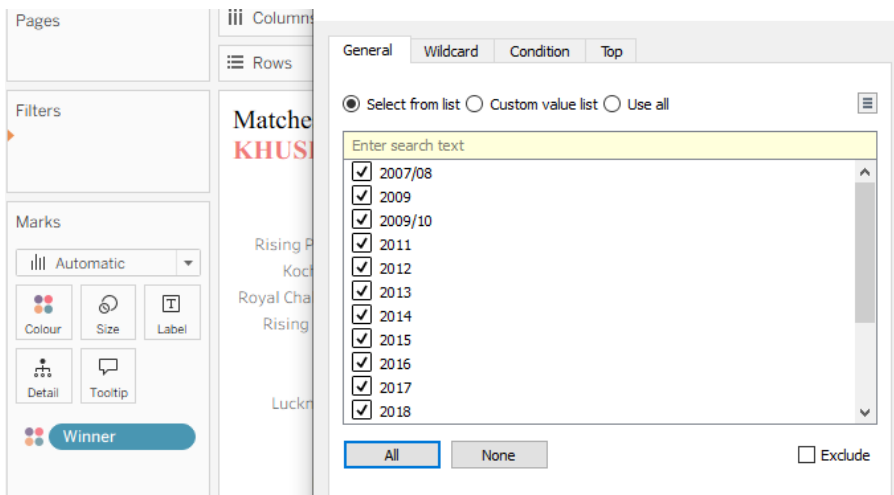
Step 4: Assemble the Dashboard Layout

1. Create a new Dashboard of all

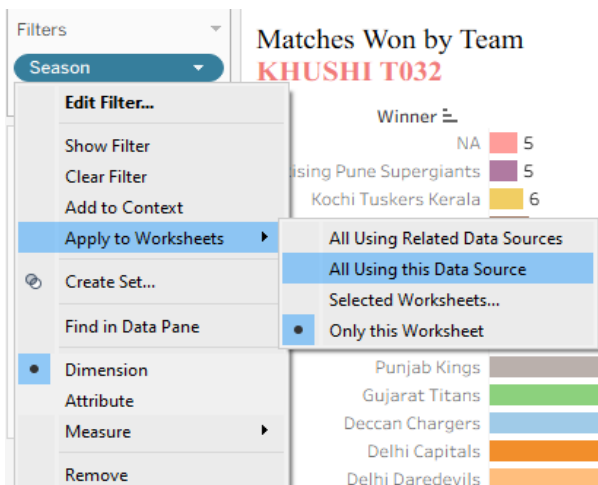
Matches Won by Team
KHUSHI T032Toss Decision Impact
KHUSHI T032Top Match Venues
KHUSHI T032**B] Add filters and interactive controls :**

Step 1: Add a Season Filter across All Worksheets

1. Go back to your Matches Won by Team worksheet.
2. Drag Season from the Data pane onto the Filters card.
3. Select All and click OK.

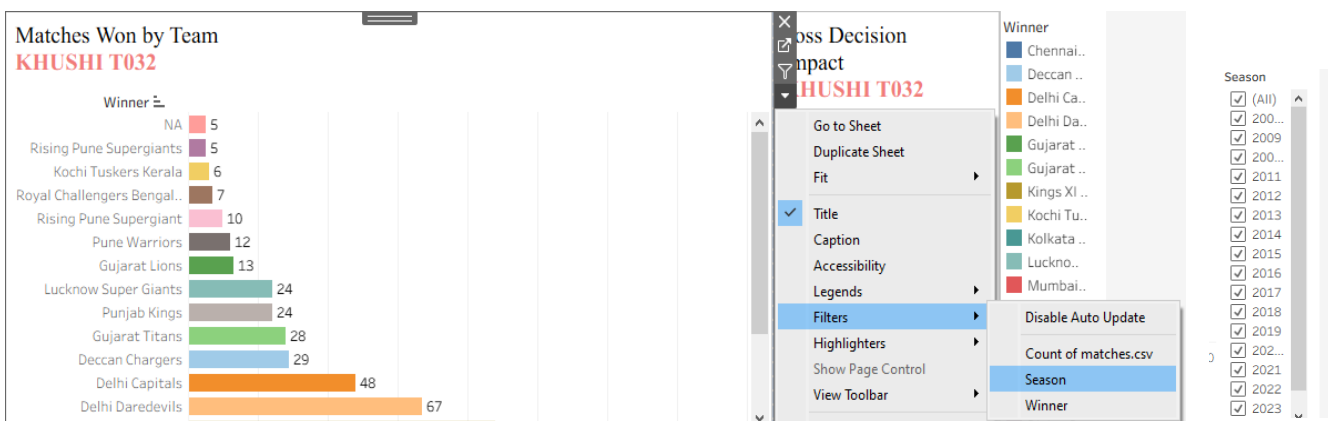


4. Right-click the Season pill on the Filters card and select Apply to Worksheets ☐ All This Data Source (This ensures changing the season updates all dashboard charts simultaneously).



Step 2: Add Filters to the Dashboard View

1. Switch back to your main Dashboard tab.
2. Click on the Matches Won by Team chart box to select it.
3. Click the small More Options drop-down arrow in the top-right corner of the chart outline box.
4. Go to Filters ☐ select Season and then The Season filter box will appear on your right sidebar.

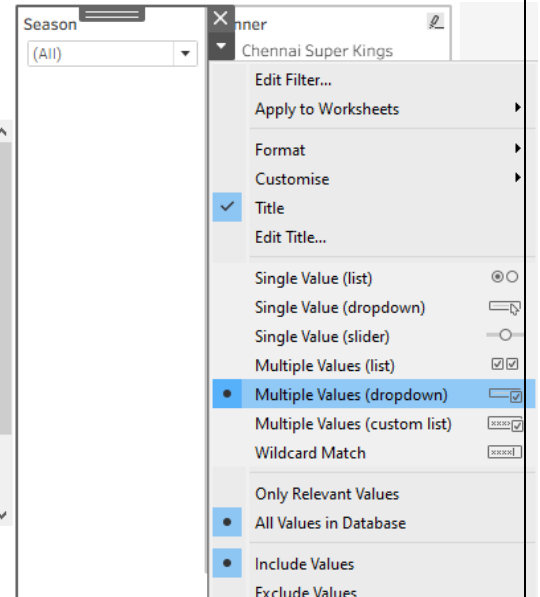
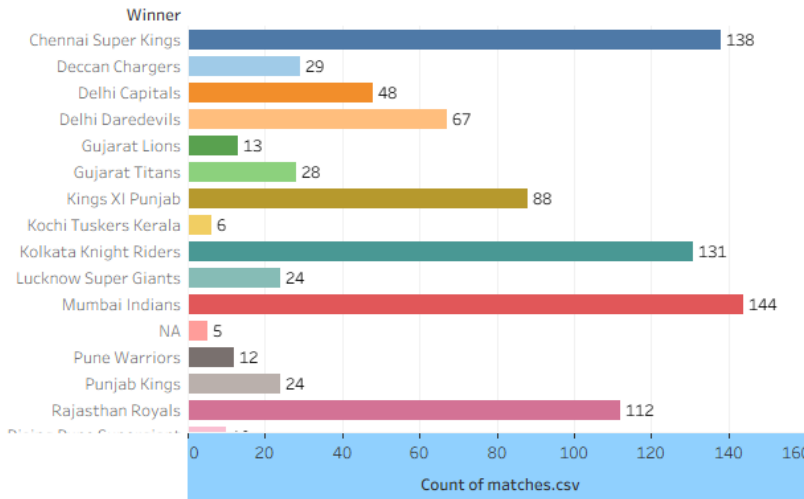


Step 3: Customize Filter Styles for User-Friendly Interactivity

1. Click the drop-down arrow on the top right of the new Season filter container box on your dashboard.
2. Change the filter style to Multiple Values (Dropdown) or Wildcard Match / Single Value List (Dropdown keeps the dashboard clean).

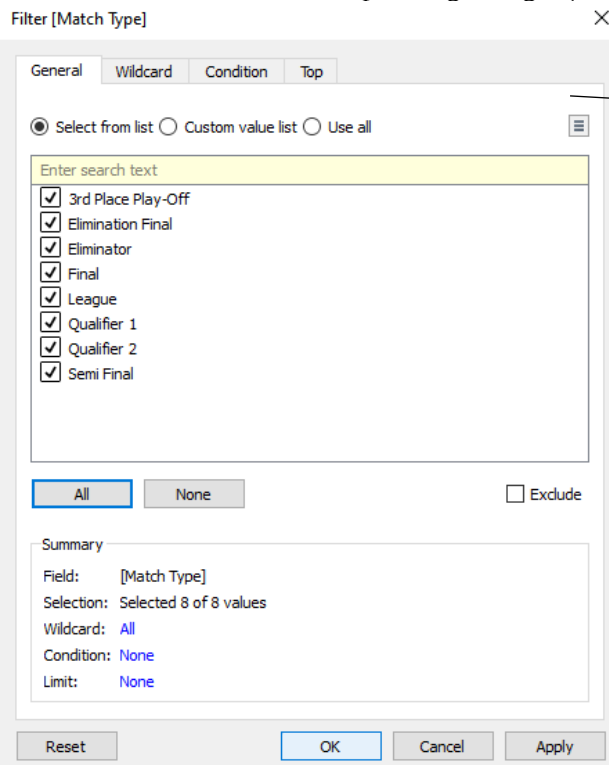
Matches Won by Team

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3. Repeat the process to add two more interactive filters:

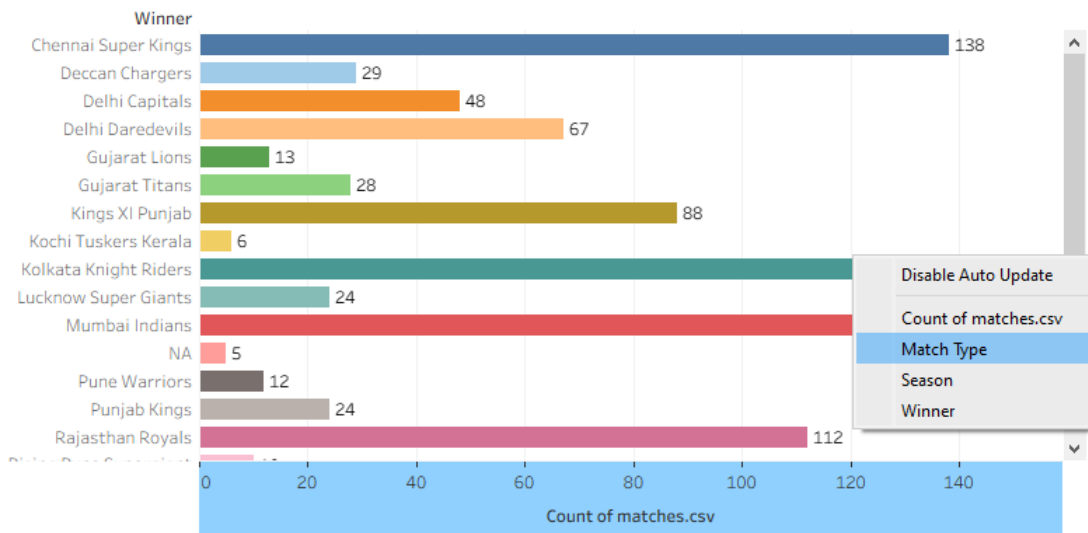
- o Click once anywhere inside your Matches Won By Team bar chart. A gray border will appear around it, indicating it is selected.
- o Click the downward-pointing triangle (More Options) located on the top right edge of that gray border.



Hover over Filters in the menu and select Match Type. (Note: If it is not listed, you will need to navigate back to your "Matches Won By Team" worksheet, drag Match Type to the Filters shelf there, and then return to your dashboard).

Matches Won by Team

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Chennai Super Kings

Deccan Chargers

Delhi Capitals

Go to Sheet

Duplicate Sheet

Fit

Title

Caption

Accessibility

Legends

Filters

Highlighters

Show Page Control

View Toolbar

Use as Filter

Ignore Actions

Add Show/Hide Button

Floating

Match Type

(All)

3rd Place Play-Off

Elimination Final

Eliminator

Final

League

Qualifier 1

Qualifier 2

Semi Final

Edit Filter...

Apply to Worksheets

Format

Customise

Title

Edit Title...

Single Value (list)

Single Value (dropdown)

Single Value (slider)

Multiple Values (list)

Multiple Values (dropdown)

Multiple Values (custom list)

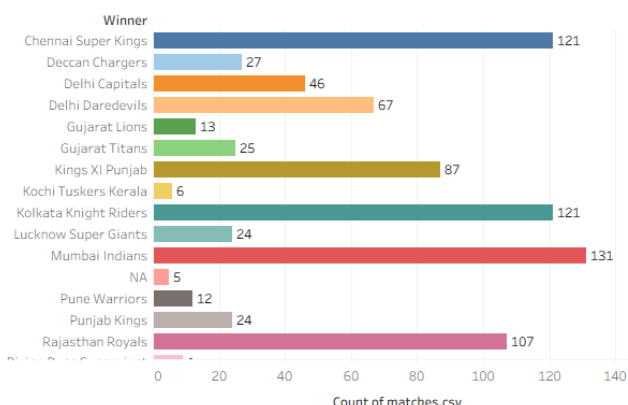
Wildcard Match

First go back to the worksheet

1. Click **Go to Sheet** from the menu.
2. You will return to the **Matches Won By Team** worksheet.
3. On the left side, find **City** in your data fields.
4. **Drag City to the Filters shelf** at the top.
5. A **Filter dialog box** will appear.
6. Click **OK**.
7. Go back to your **Dashboard**.
8. Click the **Matches Won By Team** bar chart once.
9. Click the **▼ More Options** button.
10. Select: **Filters** → **City**

Matches Won by Team

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Match Type

(All)

3rd Place Play-Off

Elimination Final

Go to Sheet

Duplicate Sheet

Fit

Title

Caption

Accessibility

Legends

Filters

Highlighters

Show Page Control

View Toolbar

Use as Filter

Ignore Actions

Add Show/Hide Button

Floating

City

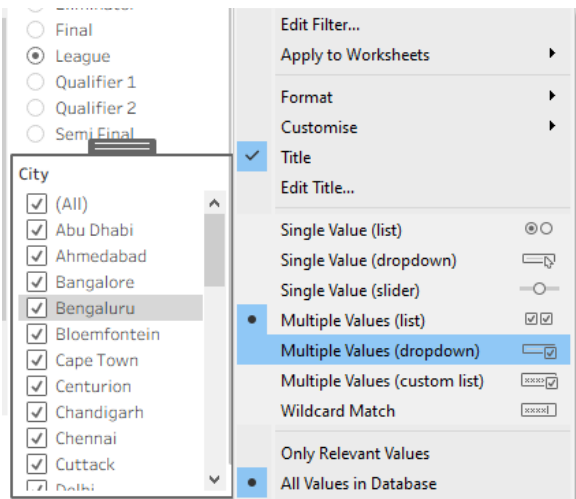
Count of matches.csv

Match Type

Season

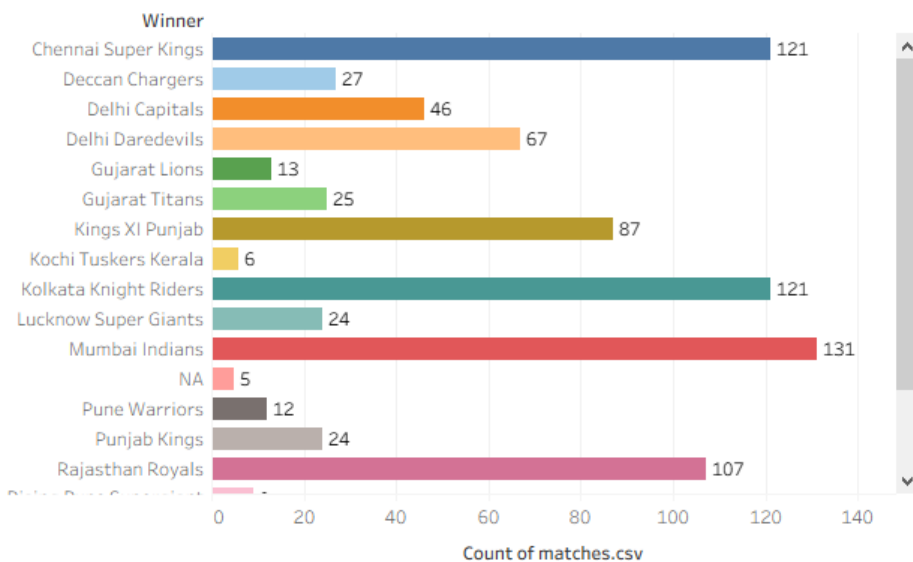
Winner

11. On the **City** filter card, click its small ▼.
12. Choose: **Multiple Values** → **Dropdown**



Matches Won by Team

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Match Type

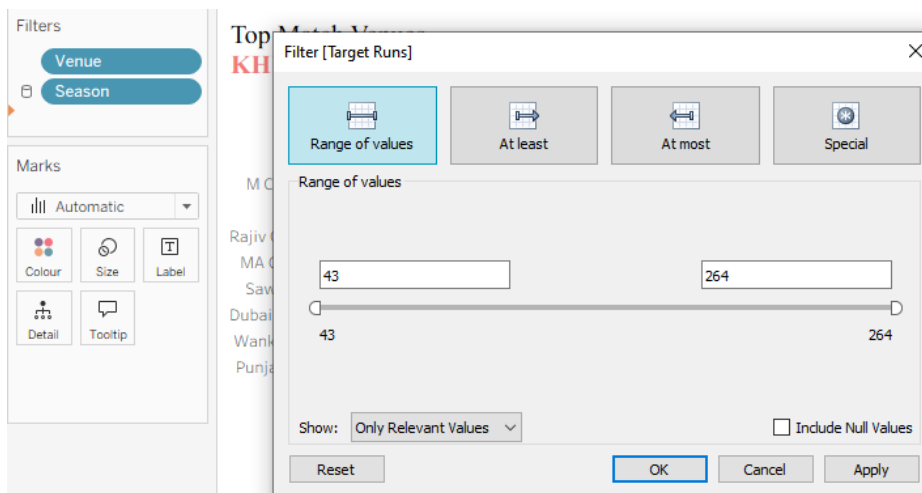
- ☒ (All)
- ☐ 3rd Place Play-Off
- ☐ Elimination Final
- ☐ Eliminator
- ☐ Final
- ☒ League
- ☐ Qualifier 1
- ☐ Qualifier 2
- ☐ Semi Final

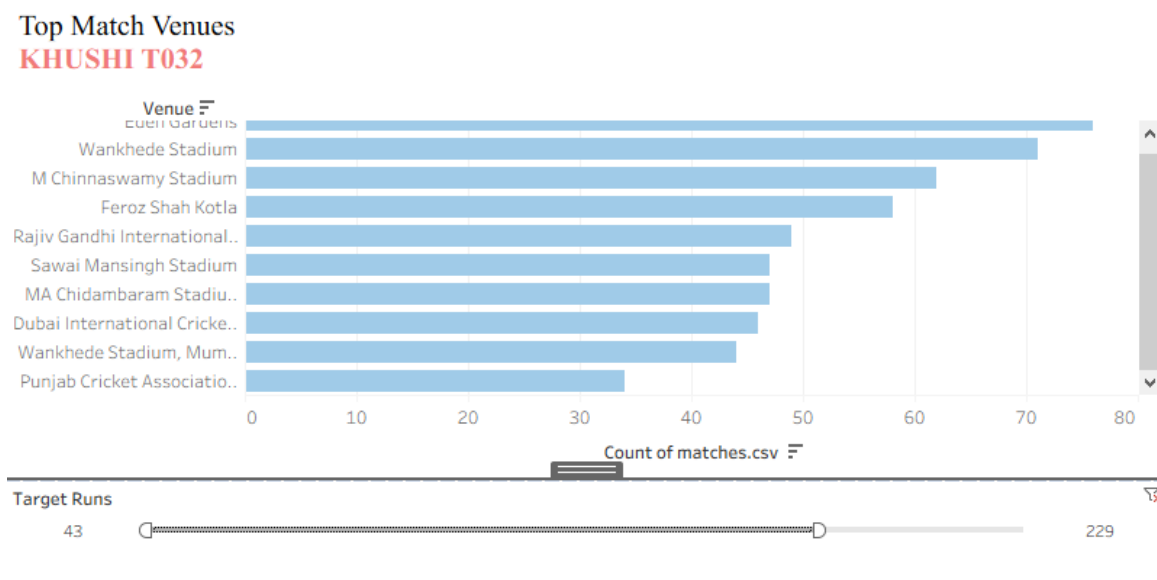
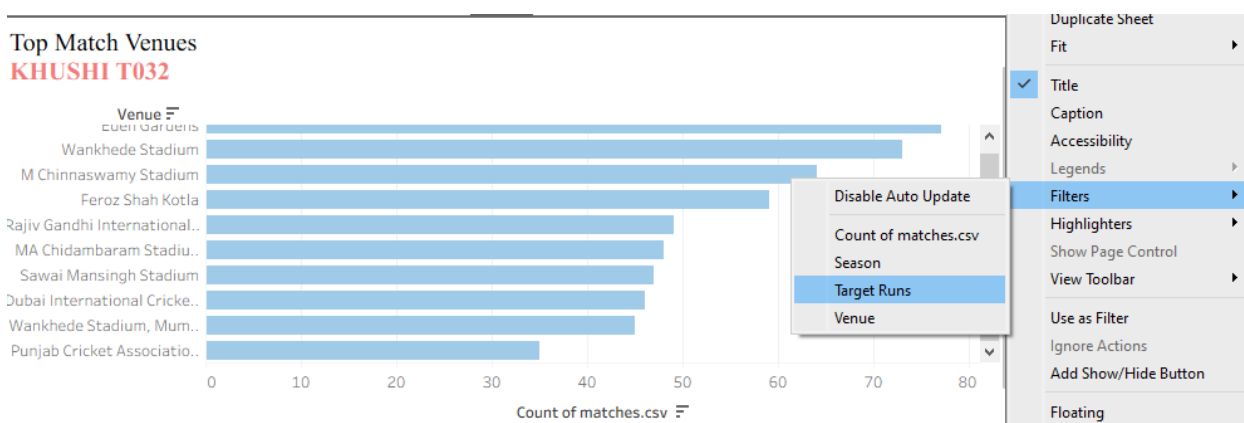
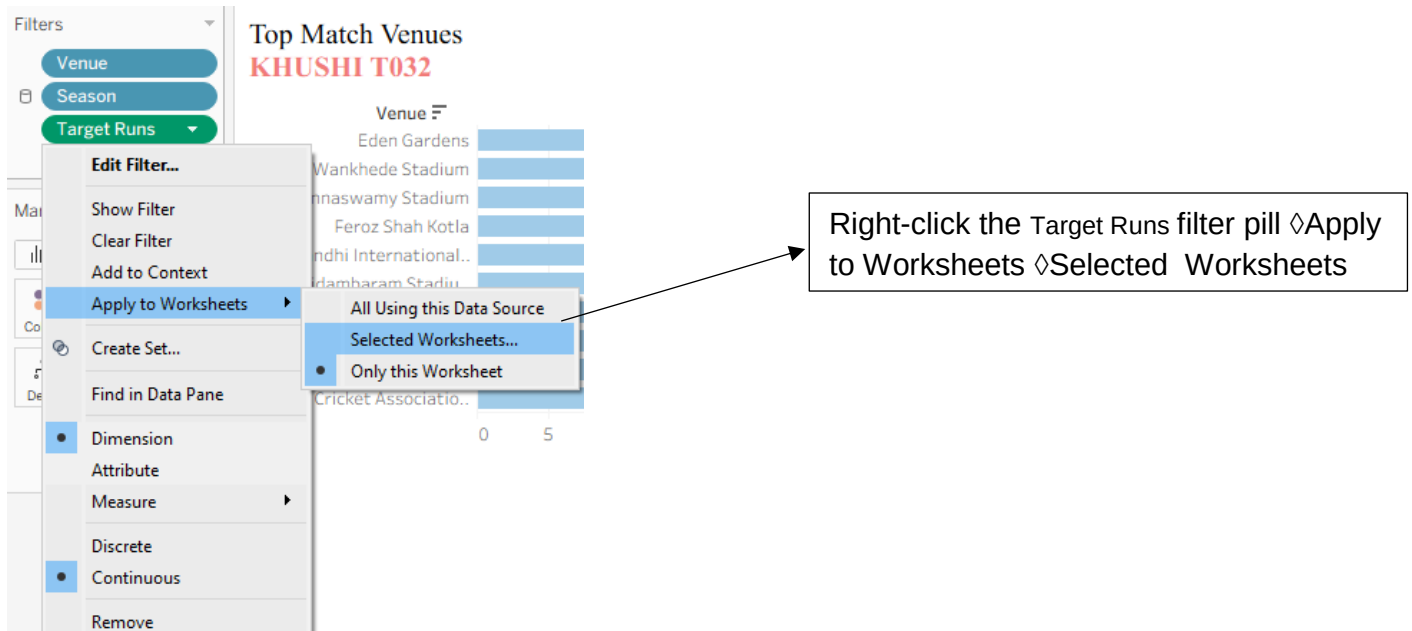
City

(All) ▼

Step 4: Add a Target Runs Range Slider (Interactive Measure Control)

1. Go to the Top Venues sheet.
2. Drag Target Runs to the Filters card.
3. Choose Range of Values and click OK.





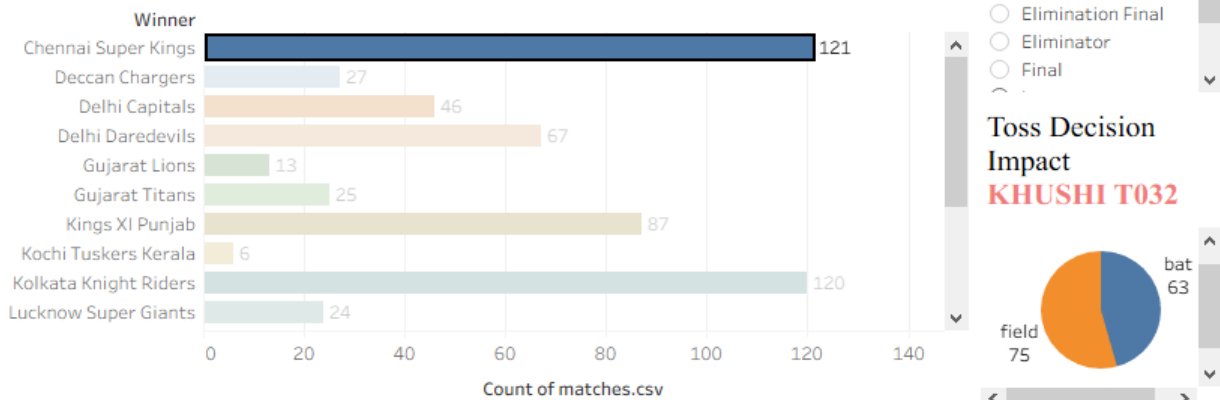
C) Implement highlight and filter actions :

1. Click on the Matches Won by Team chart box on your dashboard to select it.
- 2 Look at the small vertical toolbar that pops up on the top right side of the sheet border.
3. Click the Use as Filter icon (it looks like a small funnel symbol).

4. Test it: Click on Chennai Super Kings or Mumbai Indians in the bar chart. You will see both the Top Venues chart and the Toss Decision Impact pie chart dynamically update to show data only for that selected team! Click the selected bar again to clear the filter.

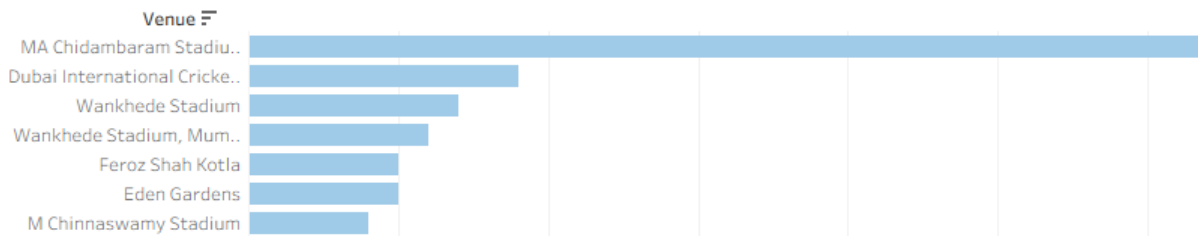
Matches Won by Team

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Top Match Venues

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Step 2: Set Up an Explicit Highlight Action via the Dashboard Menu

Next, let's create a dedicated Highlight Action using Tableau's main Action Manager:

1. On the top menu bar, go to Dashboard --> Actions...
2. In the Actions window, click Add Action > and choose Highlight...

Dashboard Story Analysis Map Format Server Window Help

New Dashboard

Device Layouts

Show Grid

Grid Options...

Format

Copy Selected Dashboard Item

Copy Image

Export Image...

Clear

Show Title

Actions... **Ctrl+Shift+D**

Auto Update

Run Update

Add Phone Layouts to Existing Dashboards

Add Phone Layouts to New Dashboards

Actions

Actions let you create interactive relationships between the dashboard and the web.

Show actions for

☒ This workbook ☐ This sheet

Name	Run On	Score
Filter...		
Highlight...		
Go to URL...		
Go to Sheet...		
Change Parameter...		
Change Set Values...		

Add Action

3. Name the action: Highlight Team Across Charts.
4. Set up the rules:

- o Source Sheets: Select Matches Won by Team.
- o Run action on: Select Hover (or Select if you prefer clicking).
- o Target Sheets: Select Top Venues and Toss Decision Impact.
- o Target Fields: Leave as All Fields or explicitly select Winner / Team1.

5. Click OK, then click OK again to save

Add Highlight Action
✕

Name

Highlight Team Across Charts
Insert ▼

Source Sheets

Dashboard 1 ▼

☒ Matches Won by Team
☐ Top Venues
☐ Toss Decision Import

Run action on

☒ Hover
☐ Select
☐ Menu

Target Sheets

Dashboard 1 ▼

☐ Matches Won by Team
☒ Top Venues
☒ Toss Decision Import

Target Highlighting

☒ All Fields
☐ Dates and Times
☐ Selected Fields

☐ Winner

Cancel

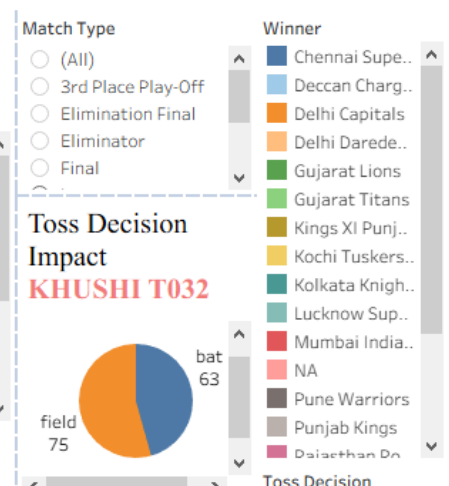
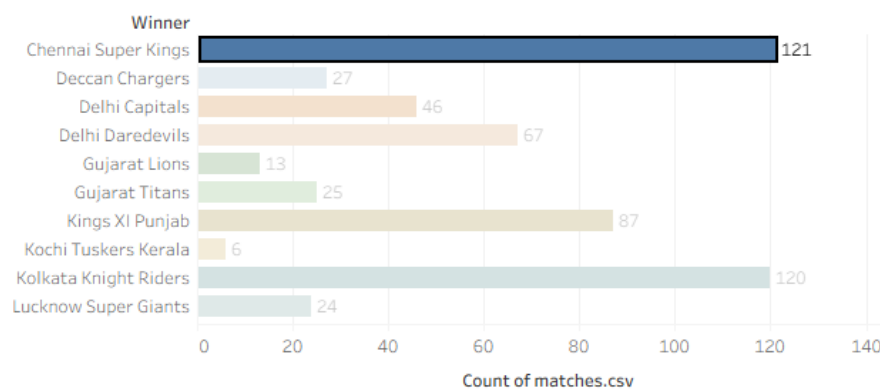
OK

Step 3: Test Your Actions

- Hovering: Hover your mouse over any bar in Matches Won by Team to see associated data highlight across other sheets.
- Clicking: Click a venue in Top Venues (if "Use as Filter" is enabled there too) to see which teams have won the most matches at that specific stadium!

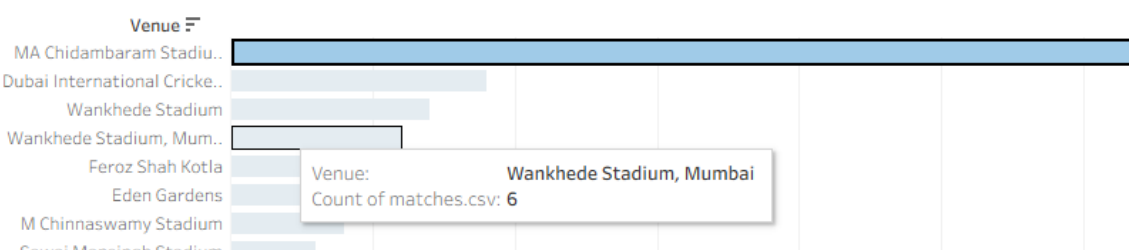
Matches Won by Team

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Top Match Venues

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Venue: Wankhede Stadium, Mumbai
Count of matches.csv: 6

Toss Decision

bat
field

City

(All) ▼

D] Use URL and set actions for interactivity :

This is where we add advanced interactivity:

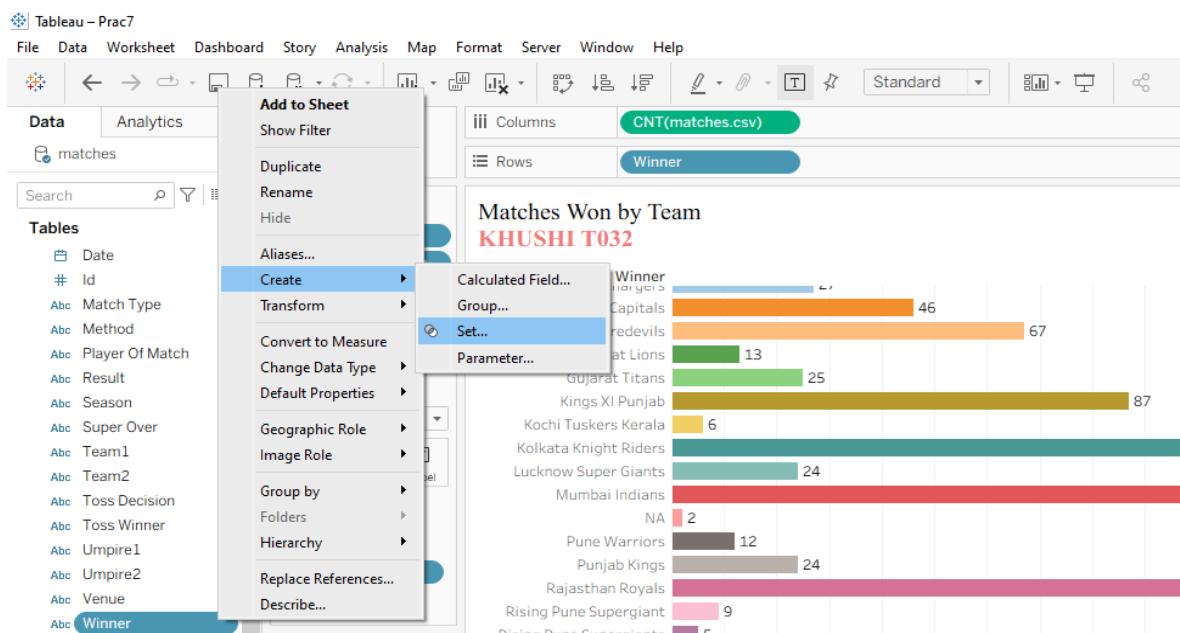
1. **URL Action:** Clicking a team or player name opens their official Wikipedia/ESPNcricinfo profile directly in a browser (or Web Page Object inside Tableau).
2. **Set Action:** Selecting one or more teams on a sheet dynamically updates a "Selected vs. Rest of League" comparison.

Task 1: Create a Set Action (Dynamic Team Comparison)

First, let's create a Set and configure a Set Action so clicking a team highlights or isolates that team against the rest of the league.

1. Create the Set in Tableau:

1. Go to the Data pane on the left side of any worksheet.
2. Right-click the Winner field $\diamond$ Create $\diamond$ Set...



Create Set

Name: Selected Team Set

General Condition Top

☒ Select from list ☐ Custom value list ☐ Use all

Enter search text

- ☐ Chennai Super Kings
- ☐ Deccan Chargers
- ☐ Delhi Capitals
- ☐ Delhi Daredevils
- ☐ Gujarat Lions
- ☐ Gujarat Titans
- ☐ Kings XI Punjab
- ☐ Kochi Tuskers Kerala
- ☐ Kolkata Knight Riders
- ☐ Lucknow Super Giants
- ☒ Mumbai Indians

All None ☐ Exclude

Summary

Field: [Winner]

Selection: Selected 1 of 20 values

Wildcard: All

Condition: None

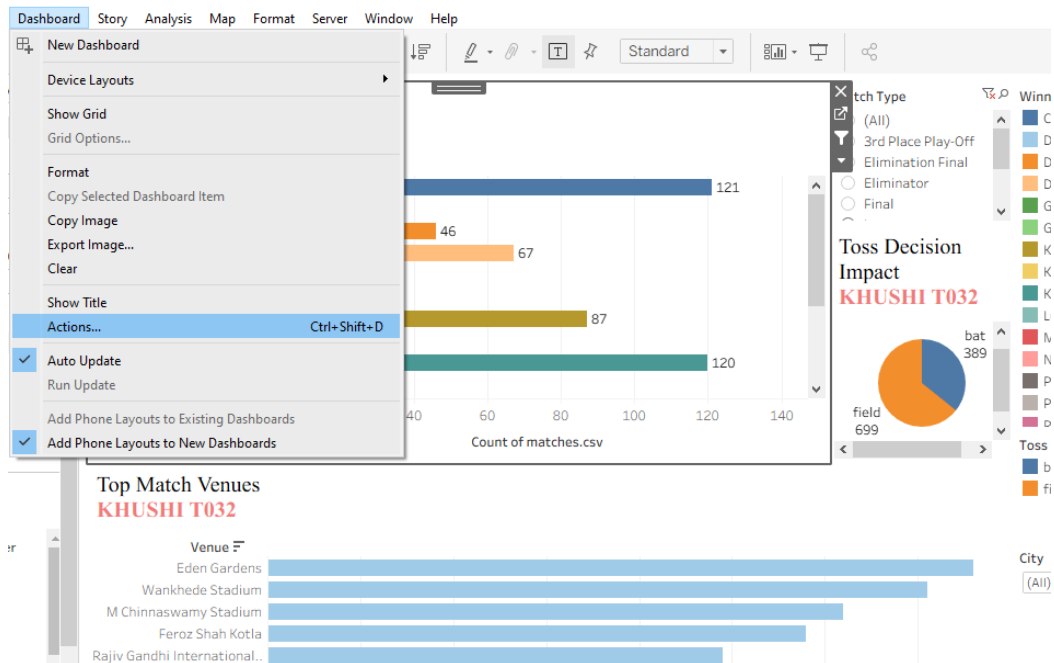
Limit: None

Reset OK Cancel

3. Name the set: Selected Teams Set.
4. Select any single team (e.g., Mumbai Indians) just as a temporary placeholder and then ok

2. Configure the Set Action on the Dashboard:

1. Return to your main Dashboard tab.
2. Go to the top menu bar: Dashboard ▾ Actions...
3. Click Add Action > ▾ select Change Set Values...



The screenshot shows the 'Actions' dialog box in Tableau. The title bar is 'Actions'. The main text says 'Actions let you create interactive relationships between data, dashboard objects, other worksheets and the web.' Below this, there are two radio buttons: 'This workbook' (selected) and 'This sheet'. A table lists actions for 'Dashboard 1 (Matches V All)'. The table has columns: Name, Run On, Source, and Fields. The 'Filter...' action is highlighted. Below the table, there are buttons for 'Add Action', 'Edit', and 'Remove'. At the bottom, there are 'Cancel' and 'OK' buttons.

Name	Run On	Source	Fields
Filter...		Dashboard 1 (Matches V All)	

4. Set up the details:

- o Name: Update Selected Team Set
- o Source Sheets: Select Matches Won by Team
- o Run action on: Select
- o Target Dataset: Choose your IPL dataset/matches file.
- o Set: Choose Selected Teams Set.
- o Running the action will: Assign values to set
- o Clearing the selection will: Remove all values from set (or Keep set values).

5. Click OK, then click OK again.

Edit Set Action



Name

Update Selected Team Set

Insert ▼

Source Sheets

Dashboard 1 ▼

☒ Matches Won by Team
☐ Top Venues
☐ Toss Decesion Import

Run action on

- ☐ Hover
☒ Select
☐ Menu
☐ Single-select only

Target Set

Selected Team Set ▼

Running the action will

- ☒ Assign values to set
☐ Add values to set
☐ Remove values from set

Clearing the selection will

- ☐ Keep set values
☐ Add all values to set
☒ Remove all values from set

Cancel

OK

Task 2: Create a URL Action (External Link to Wikipedia / ESPNcrinfo)

Now let's configure an action where clicking a team in your bar chart opens its Wikipedia page automatically.

1. Add the Dashboard Action:

- Go to the top menu: Dashboard ◇ Actions...
- Click Add Action > ◇select Go to URL..

Actions



Actions let you create interactive relationships between data, dashboard objects, other worksheets and the web.

Show actions for

- ☒ This workbook ☐ This sheet

Name	Run On	Source	Fields
Filter...	t	Dashboard 1 (Matches V All	
Highlight...	t	Dashboard 1 (Matches V Selected Team Set	
Go to URL...			
Go to Sheet...			
Change Parameter...			
Change Set Values...			

Add Action ▼

Edit

Remove

Cancel

OK

3. Configure the URL Action:

- o Name: Open Team Wikipedia Page
- o Source Sheets: Select Matches Won by Team
- o Run action on: Menu (This displays a clean clickable hyperlink inside the tooltip when a user clicks a bar, preventing accidental page loads).
- o URL: Paste the following dynamic URL: <https://en.wikipedia.org/wiki/<Winner>>

(Note: Click the > button next to the URL input box to insert the <Winner> field dynamically).

Add URL Action

Name

Open Team Wikipedia Page
Insert

Source Sheets

Dashboard 1

☒ Matches Won by Team
☐ Top Venues
☐ Toss Decision Import

Run action on

☐ Hover
☐ Select
☒ Menu

URL Target

☒ New Tab if no Web Page Object Exists
☐ New Browser Tab

URL

https://en.wikipedia.org/wiki/<Winner>
Insert

https://en.wikipedia.org/wiki/Chennai_Super_Kings

> Data Values

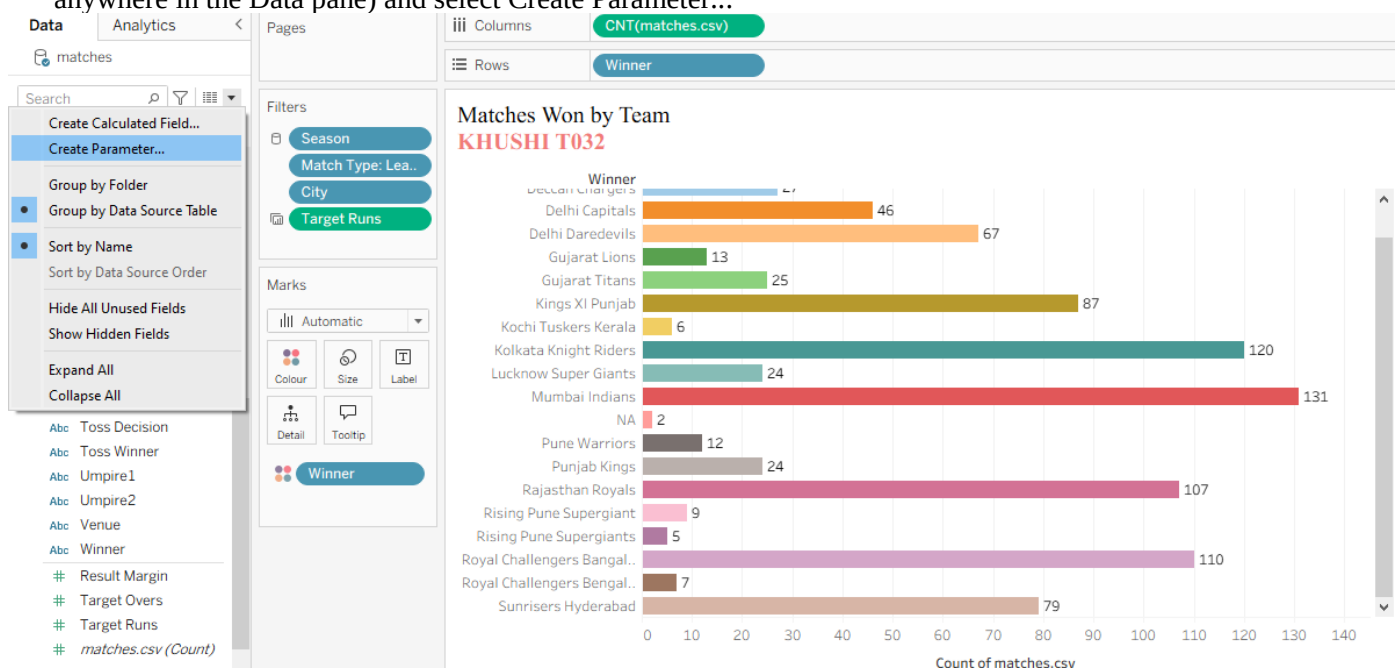
Cancel
OK

E]Create dynamic dashboards using parameter :

Parameters allow users to control what dimensions or measures are shown on a chart. Instead of creating separate sheets for "Wins by Season," "Wins by City," or "Wins by Toss Decision," we will create a single dynamic chart where the user can swap dimensions on the fly using a Parameter.

Step 1: Create a Parameter for Dimension Swapping

1. Go to any worksheet (e.g., Matches Won by Team).
2. At the top of the Data pane on the left, click the small drop-down arrow next to the search bar (or right-click anywhere in the Data pane) and select Create Parameter...



Create Parameter

Name: Select Dimensions

Properties:

Data type: String | Display format: Winner

Current value: Winner | Value when workbook opens: Current value

Allowable values:

☐ All ☒ List ☐ Range

Value	Display As
Winner	Winner
City	City
Toss Winner	Toss Winner
Venue	Venue

☒ Fixed ☐ When workbook opens

Add values from ▼

Remove Selected

Cancel OK

3. Configure the parameter options:

- o Name: Select Dimension
- o Data type: String
- o Allowable values: List
- o List of values:

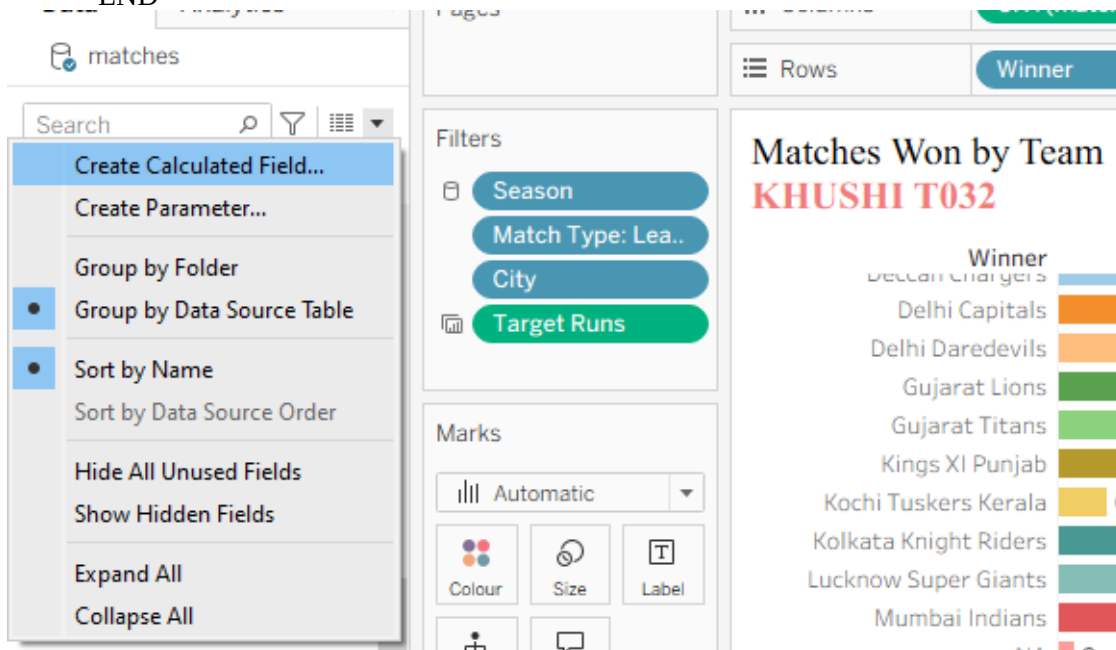
Add the following exact names in the Value column: Winner , City , Toss Winner ,Venue

Step 2: Create a Calculated Field to Link the Parameter

A parameter doesn't do anything on its own, we need a calculated field to tell Tableau how to respond when the parameter value changes.

1. Right-click in the Data pane ◇ Create Calculated Field...
2. Name the field: Dynamic Dimension
3. Enter the following formula:

```
CASE [Select Dimension]
WHEN 'Winner' THEN [Winner]
WHEN 'City' THEN [City]
WHEN 'Toss Winner' THEN [Toss Winner]
WHEN 'Venue' THEN [Venue]
END
```



Dynamic Dimension



```

CASE [Select Dimensions ]
  WHEN 'Winner' THEN [Winner]
  WHEN 'City' THEN [City]
  WHEN 'Toss Winner' THEN [Toss Winner]
  WHEN 'Venue' THEN [Venue]
END

```

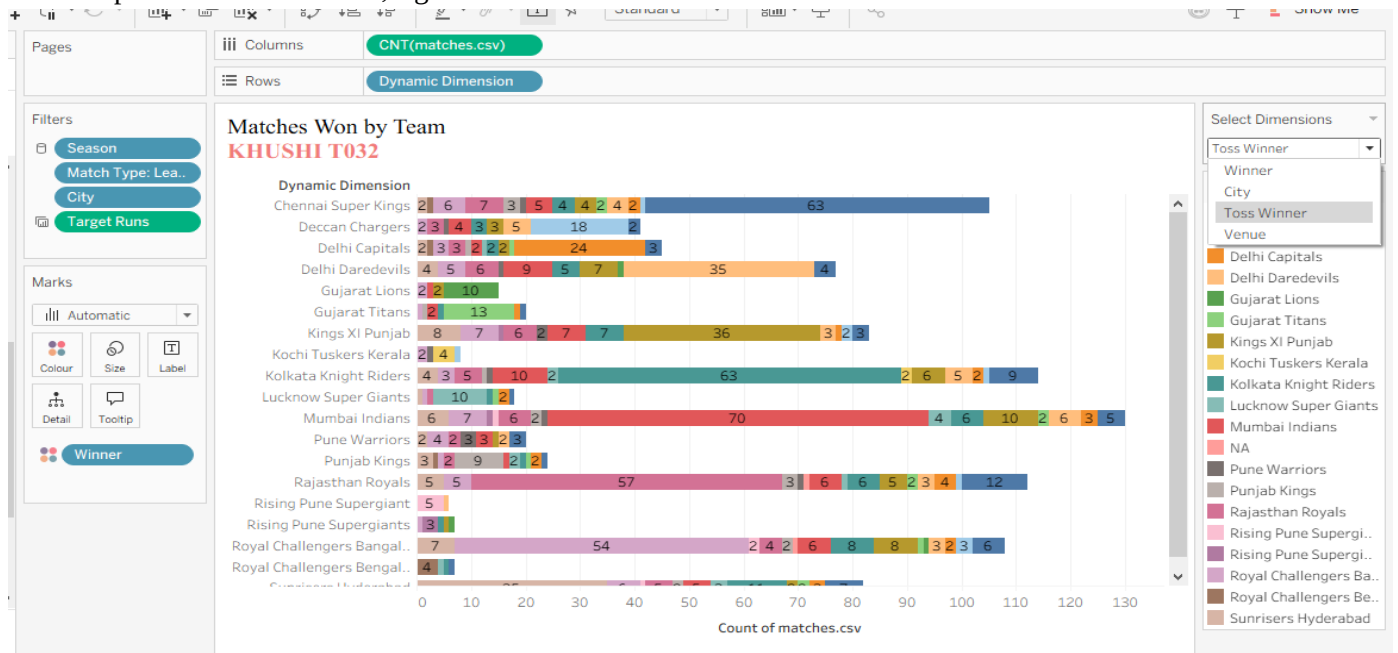
The calculation is valid.

Apply

OK

Step 3: Replace the Fixed Field with the Dynamic Field

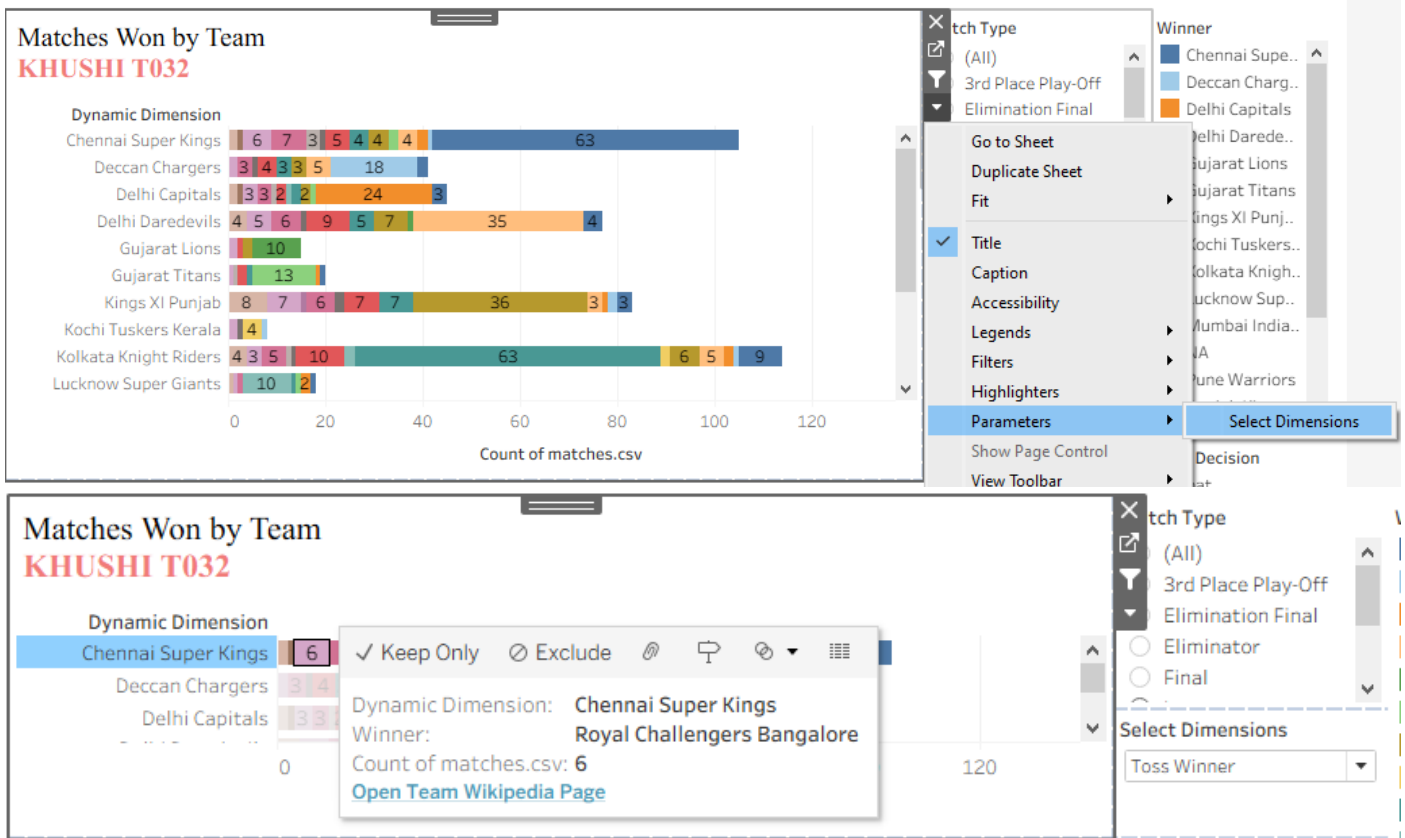
1. On your Matches Won by Team sheet, locate the field currently on the Rows shelf (e.g., Winner).
2. Drag Dynamic Dimension from the Data pane and drop it directly over Winner on the Rows shelf to swap them. In the Data pane under Parameters, right-click Select Dimension and check Show Parameter.



4. Test changing the parameter selection on the right panel, you'll see the chart instantly switch between listing Teams, Cities, Toss Winners, and Venues!

Step 4: Display the Parameter Control on the Dashboard

1. Switch back to your main Dashboard tab.
2. Select the dynamic chart box.
3. Click the More Options drop-down arrow in the top right corner of the chart outline box.
4. Go to Parameters > select Select Dimension.

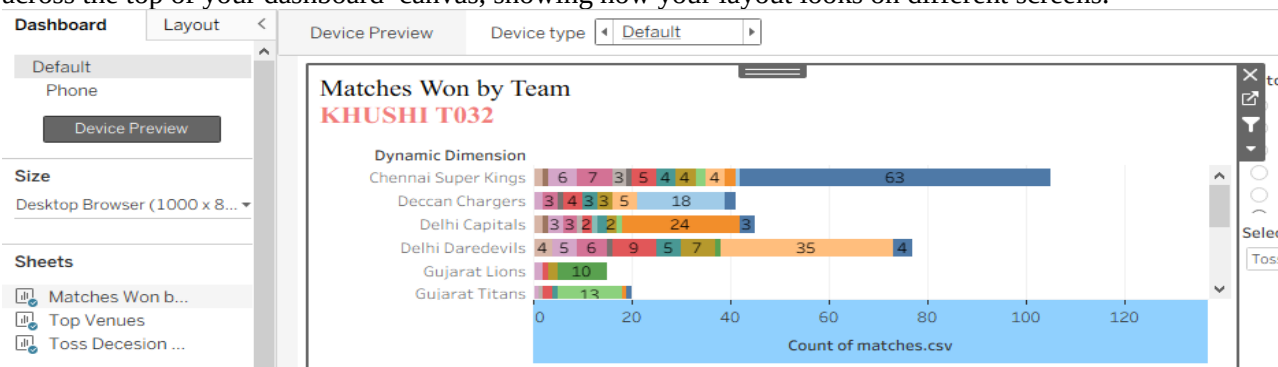


F]Design dashboards for multiple devices :

In this part, we will use Tableau's Device Designer to ensure your IPL Dashboard automatically adapts its layout when viewed on a Desktop, Tablet, or Mobile Phone.

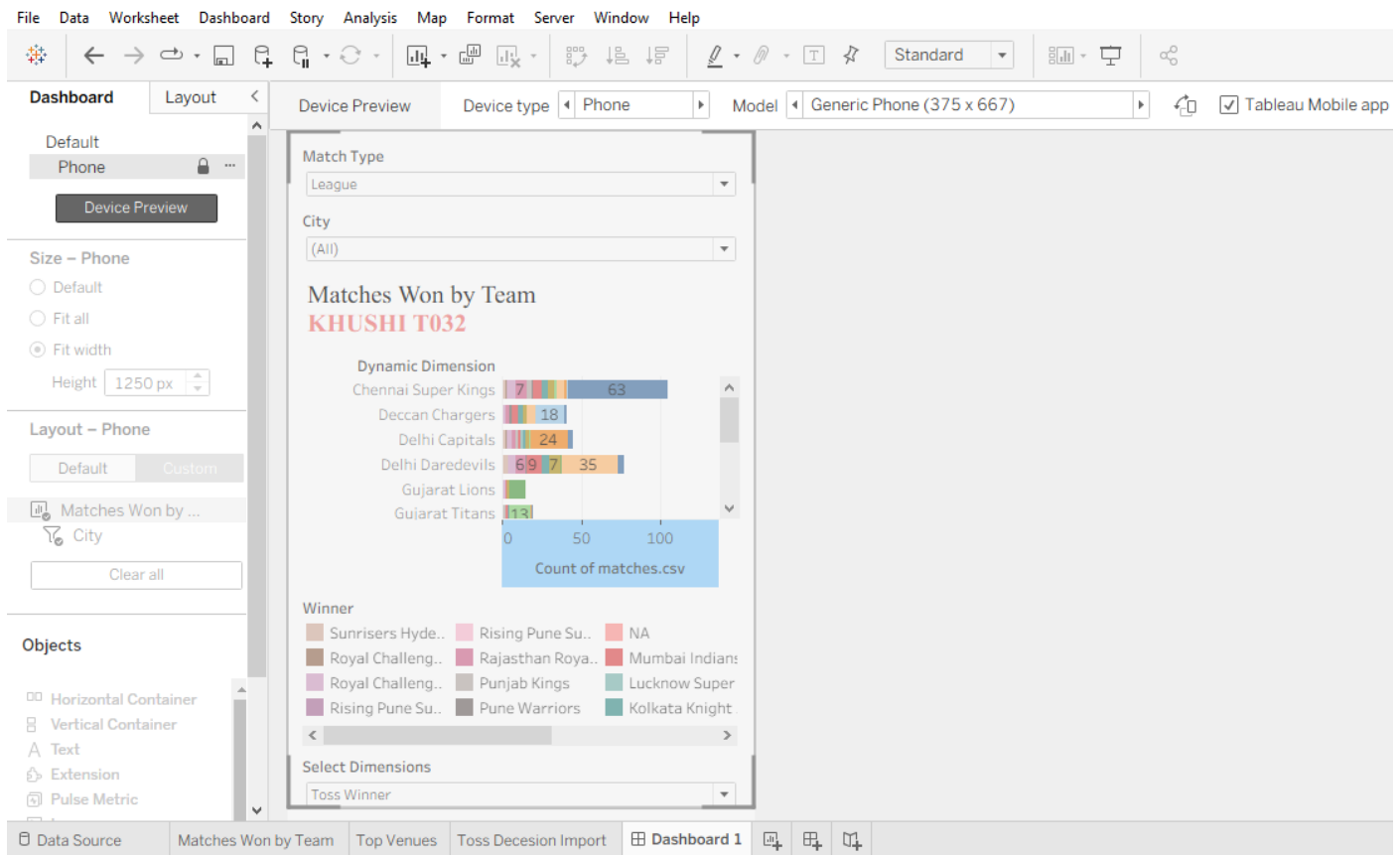
Step 1: Open the Device Designer Panel

1. Switch to your main Dashboard tab.
2. In the top-left Dashboard pane, click the Device Preview button.
3. A device preview frame and toolbar will appear across the top of your dashboard canvas, showing how your layout looks on different screens.



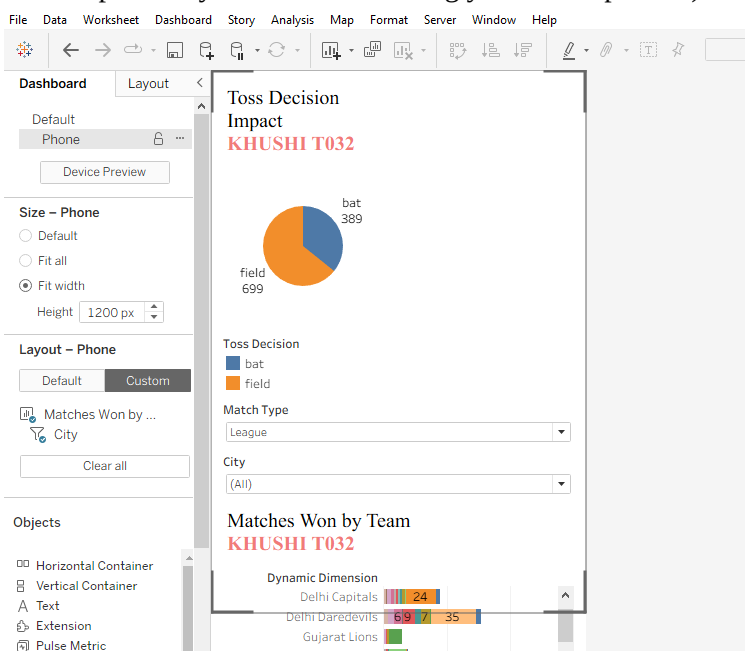
Step 2: Add a Phone Layout

1. In the top device preview bar, select Device type: Phone.
2. Click the Add Phone Layout button on the right side of the preview bar.
3. You will now see a new sub-tab named Phone appear under the Dashboard panel on the left.

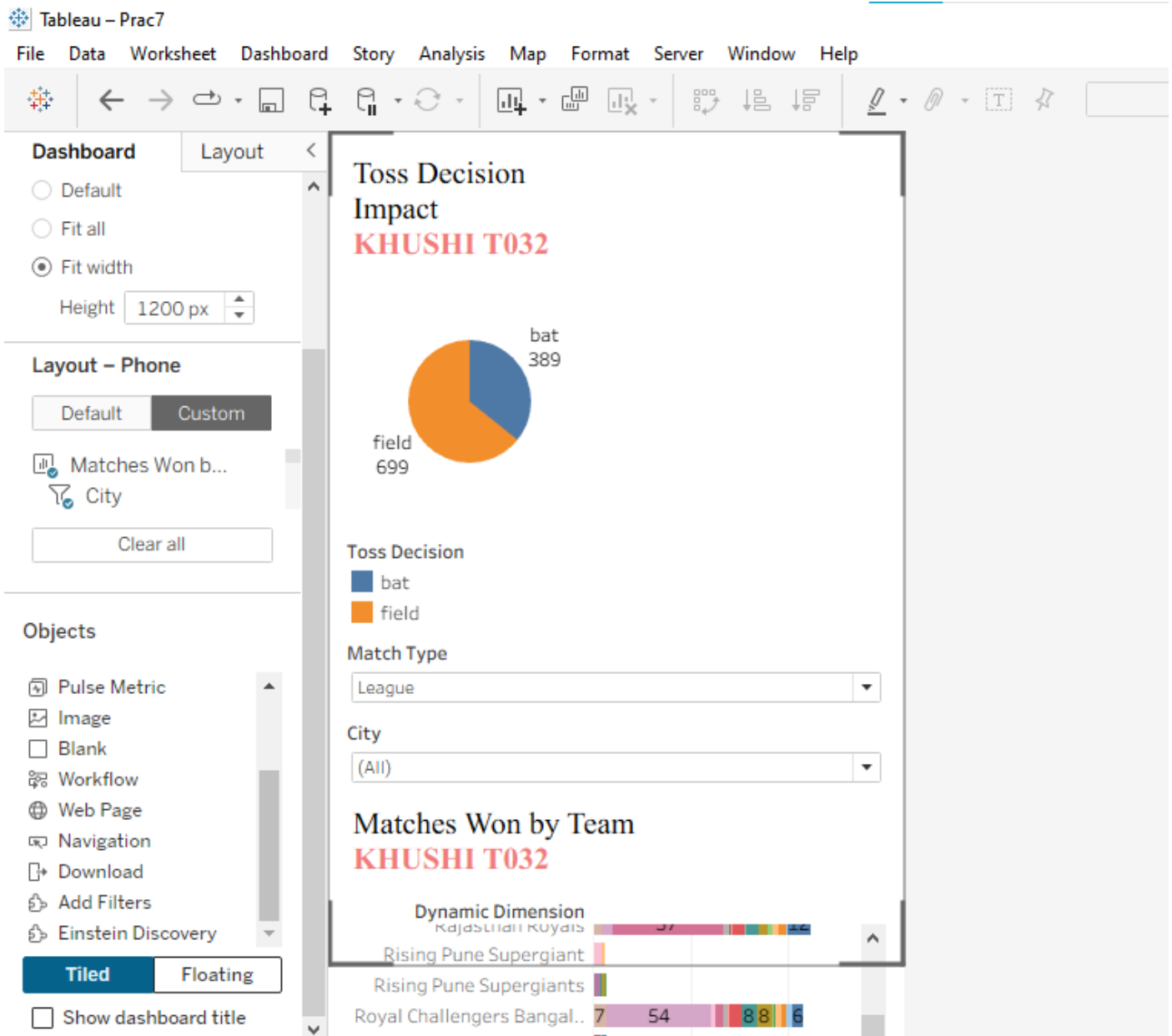


Step 3: Customize the Phone Layout for Vertical Scrolling Mobile screens require a vertical stack rather than side-by-side tiled views:

1. In the left panel under the Phone layout section, change Layout from Default to Custom (this allows you to edit the phone layout without affecting your desktop view).

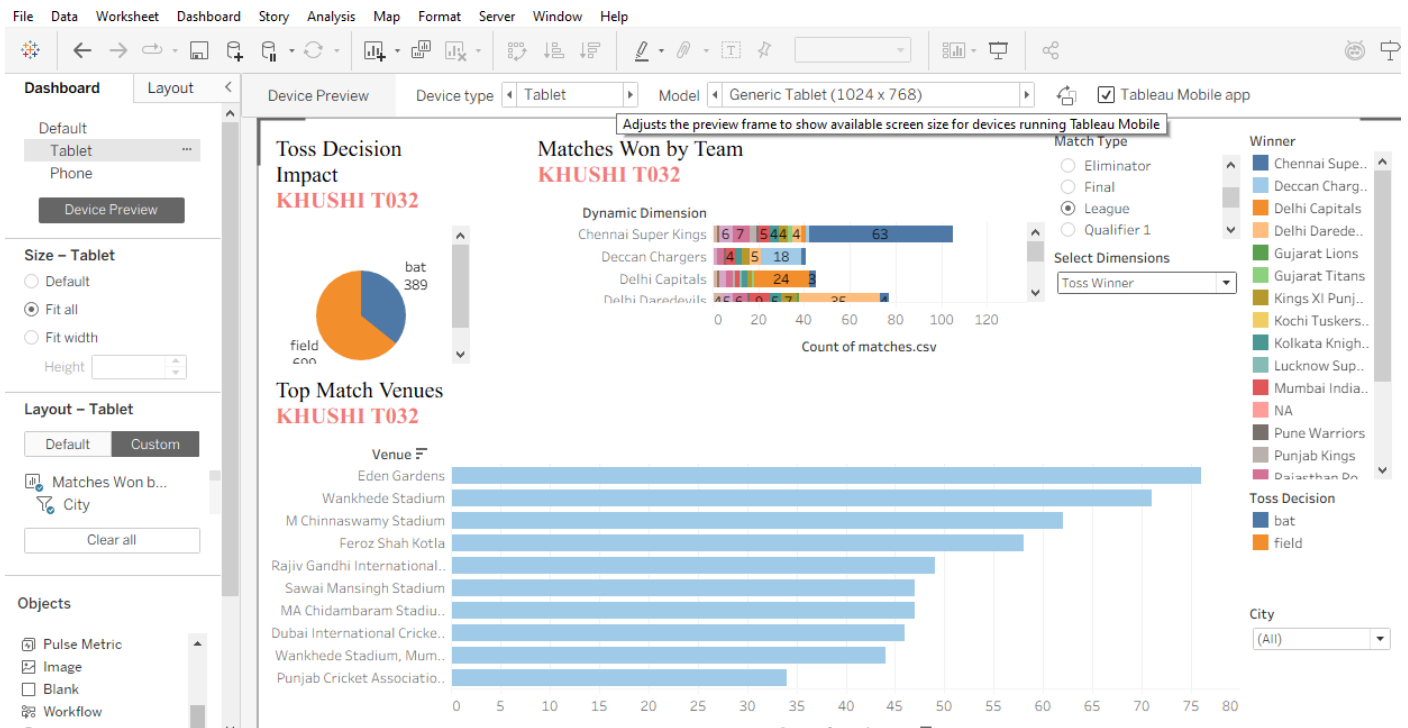


2. Tableau will automatically generate a single-column stack. Arrange the items vertically in logical order:
 - o Title & Global Filters (Season, Dynamic Dimension parameter) at the top.
 - o Matches Won by Team (Primary high-level chart).
 - o Top Venues chart below it.
 - o Toss Decision Impact pie chart at the bottom.
3. Hide secondary elements if needed: Right-click any non-essential filter or legend container in the phone tree view and select Remove from Layout (this removes it only from the phone screen, keeping your mobile dashboard clean and clutter-free).



Step 4: Add a Tablet Layout (Optional)

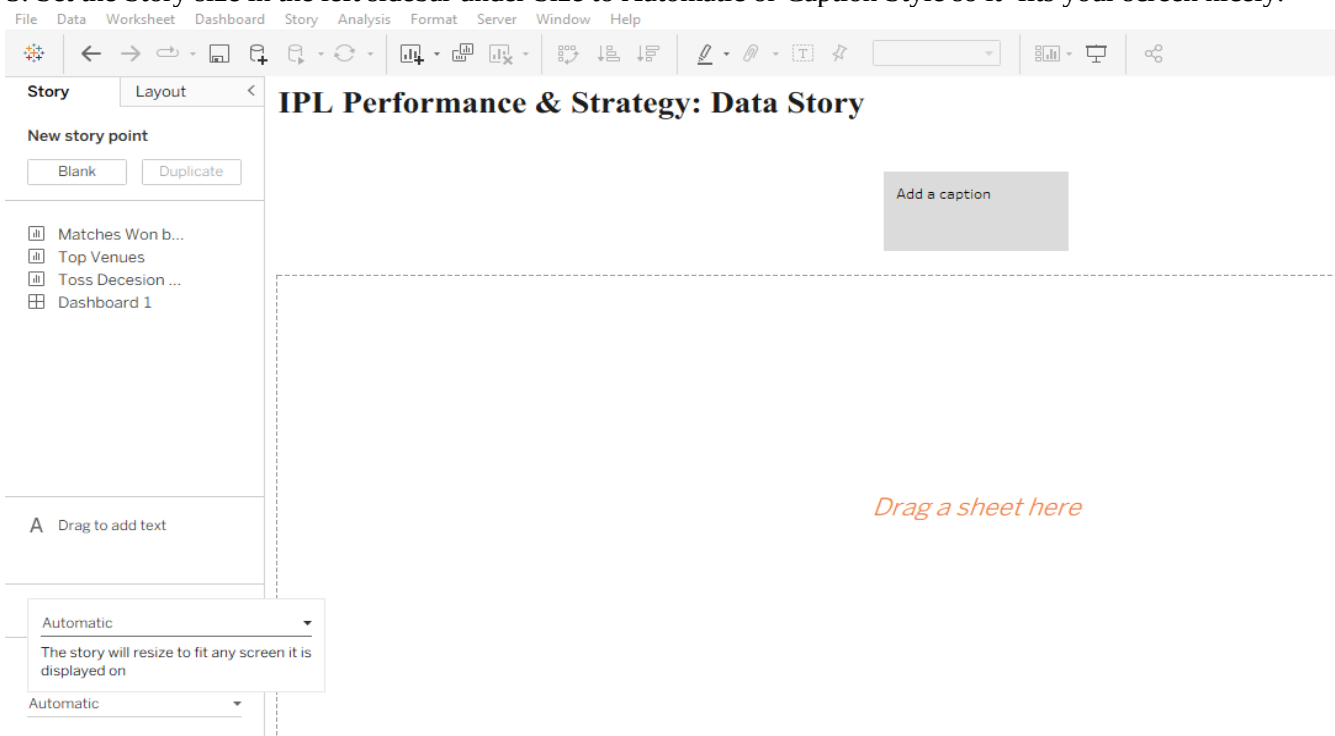
1. Change Device type in the top bar to Tablet.
2. Click Add Tablet Layout.
3. Set the layout to Custom and adjust the grid to a 2x2 matrix so all key metrics fit comfortably on an iPad/tablet screen size.



G] Build a complete data story using Story feature :

Step 1: Create a New Story Workspace

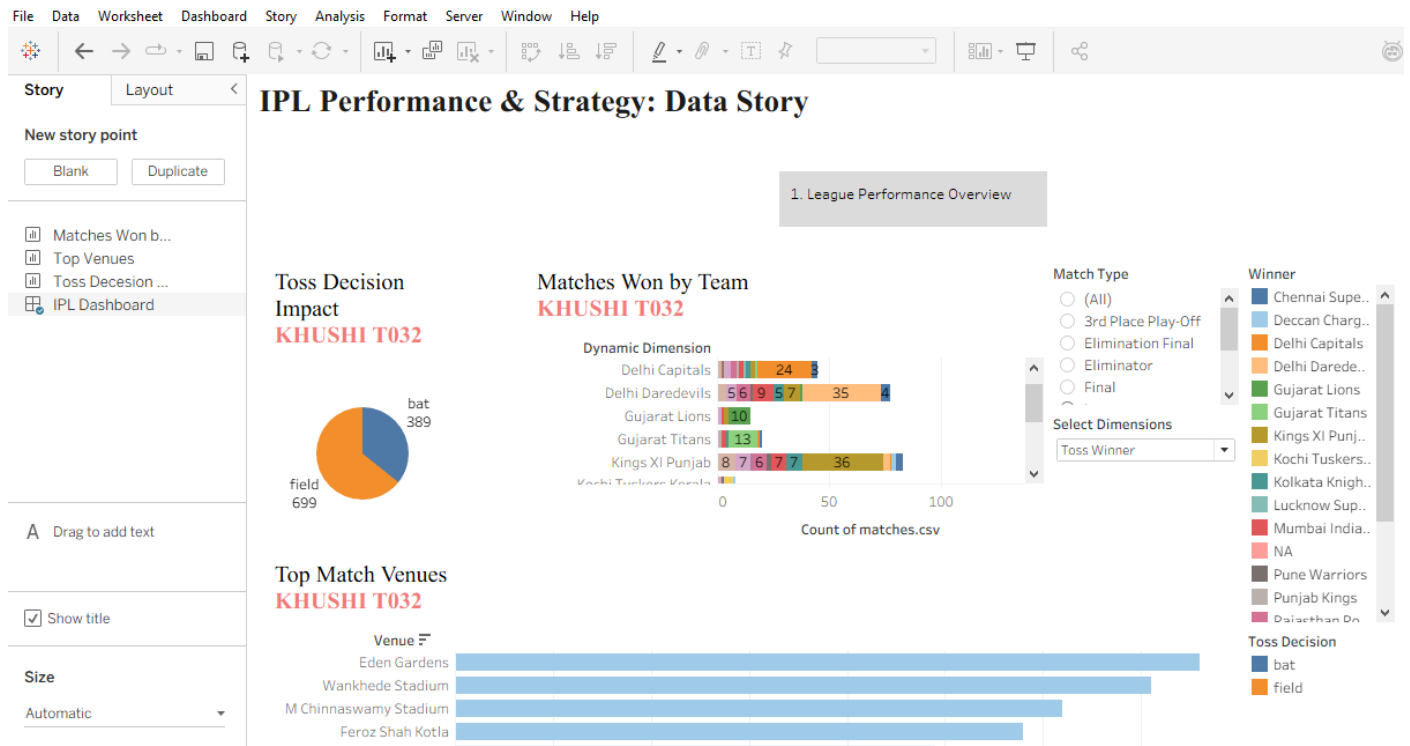
1. At the bottom of your Tableau workspace (where your Sheet and Dashboard tabs are), click the New Story tab icon (the notebook icon with a plus sign at the far right).
2. Double-click the title heading at the top of the canvas and rename your story to: IPL Performance & Strategy: Data Story
3. Set the Story size in the left sidebar under Size to Automatic or Caption Style so it fits your screen nicely.



Step 2: Story Point 1 — Macro League Overview

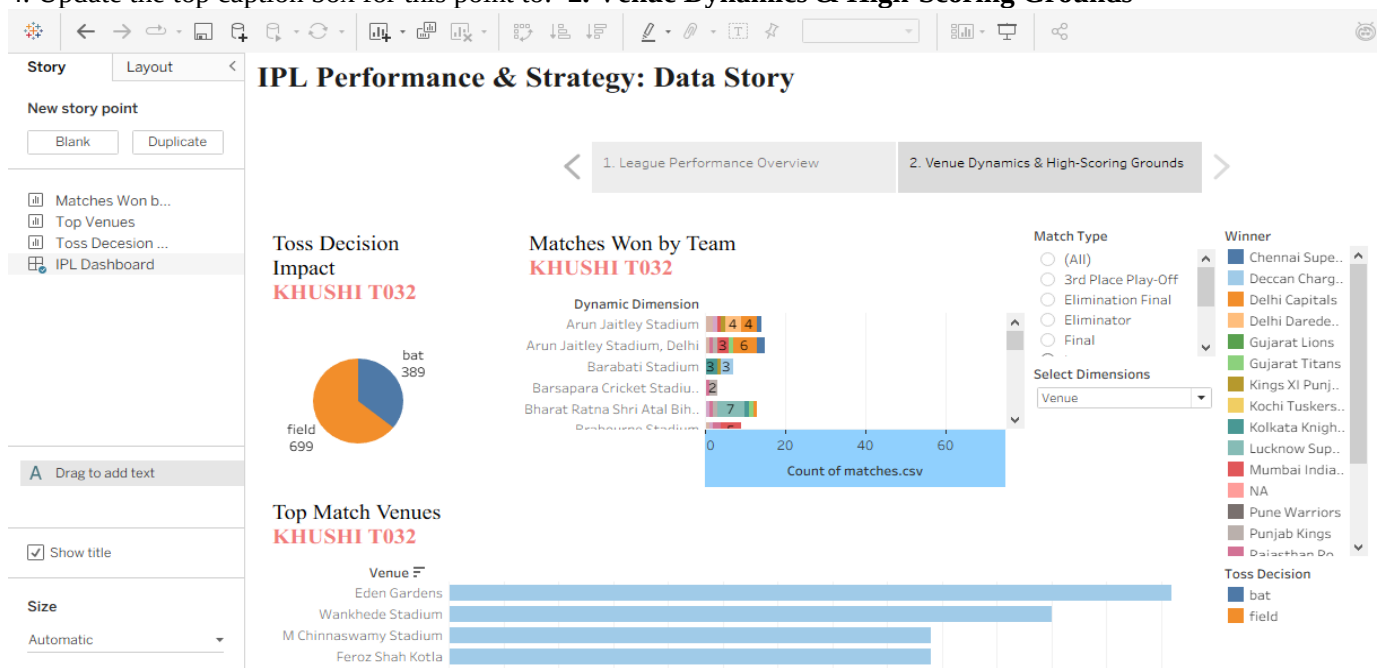
1. From the left Story pane, drag your main IPL Dashboard onto the blank canvas.

- Click inside the grey caption box above the dashboard slide (labeled "Write description text here") and type: **1. League Performance Overview**
- On this slide, keep your filters set to All Seasons so it serves as an high-level executive summary of team wins and total match dynamics across the IPL.



Step 3: Story Point 2 — Venue Analysis & Scoring Trends

- Click Blank under the New Story Point section on the left sidebar to add a second slide.
- Drag your IPL Dashboard onto the canvas again.
- In this slide, use your new Select Dimension parameter dropdown to change the primary chart view to Venue, and set your Target Runs slider to focus on high-scoring matches (180+ runs).
- Update the top caption box for this point to: **2. Venue Dynamics & High-Scoring Grounds**

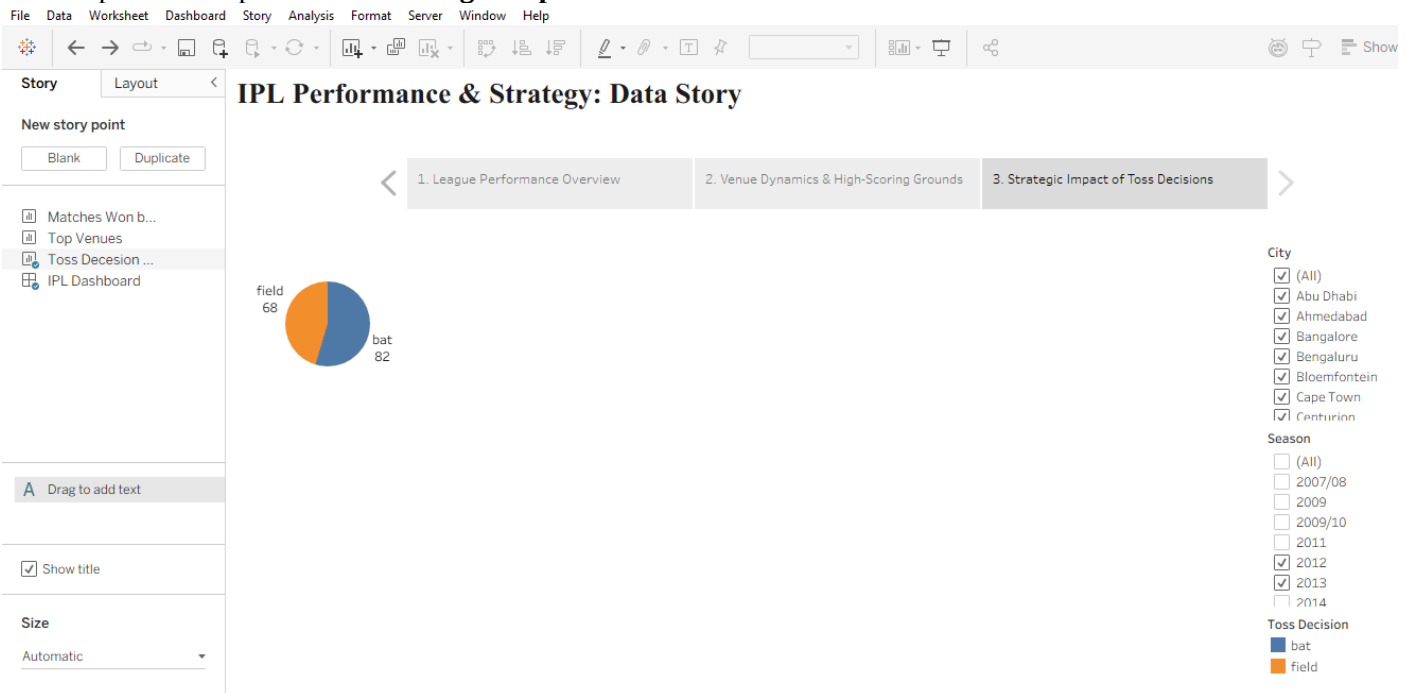


Step 4: Story Point 3 — The Impact of the Toss

- Click Blank to add a third story point.

2. Drag the Toss Decision Impact worksheet (or the full dashboard) onto this slide. 3. Use the Season or City filter controls to highlight how winning the toss and choosing to bat vs. field impacts match outcomes in specific conditions.

4. Update the caption to: **3. Strategic Impact of Toss Decisions**

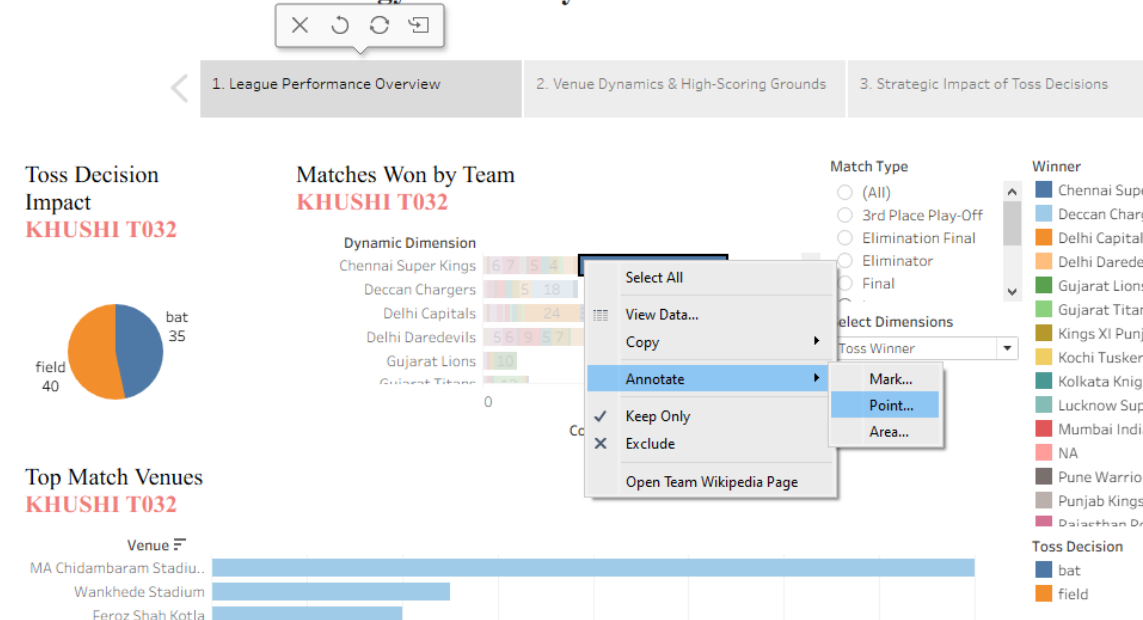


Step 5: Add Callout Annotations for Key Takeaways

To make your story presentation self-explanatory for stakeholders:

1. Go to Story Point 1 or Story Point 2.
2. Right-click a notable mark (e.g., the top team bar for Mumbai Indians or M. Chinnaswamy Stadium).
3. Select Annotate --> Mark... (or Point...).

IPL Performance & Strategy: Data Story



4. Enter a succinct takeaway (e.g., "Mumbai Indians lead total match wins across seasons").
5. Drag the callout text box to place it clearly beside the visual element.

IPL Performance & Strategy: Data Story



1. League Performance Overview

2. Venue Dynamics & High-Scoring Grounds

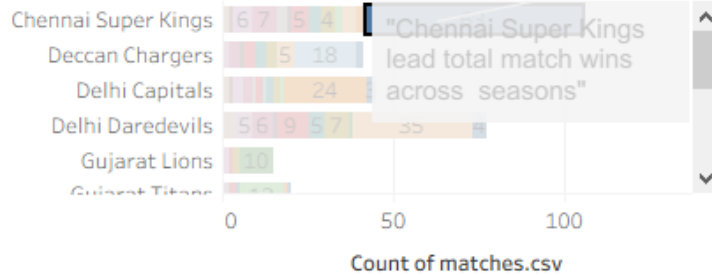
3. Strategic

Toss Decision Impact KHUSHI T032



Matches Won by Team KHUSHI T032

Dynamic Dimension



Match Type

- ☐ (All)
☐ 3rd Place Pl
☐ Elimination
☐ Eliminator
☐ Final

Select Dimension

Toss Winner

Step 6: Presentation Mode Walkthrough

1. Press F5 (or click the Presentation Mode icon in the top toolbar) to view your story full-screen.
2. Click through the story navigator buttons at the top to ensure each slide maintains its customized filter state and annotations while preserving full interactivity.

IPL Performance & Strategy: Data Story



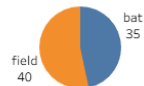
1. League Performance Overview

2. Venue Dynamics & High-Scoring Grounds

3. Strategic Impact of Toss Decisions

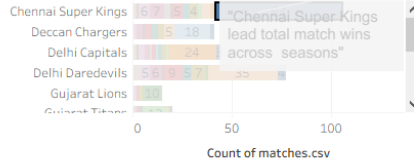


Toss Decision Impact KHUSHI T032



Matches Won by Team KHUSHI T032

Dynamic Dimension



Match Type

- ☐ (All)
☐ 3rd Place Play-Off
☐ Elimination Final
☐ Eliminator
☐ Final

Select Dimensions

Toss Winner

Winner

- ☒ Chennai Supe..
☐ Deccan Charg..
☐ Delhi Capitals
☐ Delhi Darede..
☐ Gujarat Lions
☐ Gujarat Titans
☐ Kings XI Punj..
☐ Kochi Tuskers..
☐ Kolkata Knigh..
☐ Lucknow Sup..
☐ Mumbai India..
☐ NA
☐ Pune Warriors
☐ Punjab Kings
☐ Rajasthan Ri

Toss Decision

- ☒ bat
☐ field

City

(All)

Top Match Venues KHUSHI T032

